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Problem set: *L02 — Class 2: Introduction to statistical learning (textbook exercises)*

Solutions and rubric

Instructions. These are the textbook exercises listed under *Practice* for Class 2 in `Map/class-plan.md`, grouped by book in the order of the class plan: ISLP (Block 1), Wooldridge (Block 2), Morettin & Singer (Block 3). Each statement is reproduced verbatim from the book PDF, with a source line; the Portuguese statements of Morettin & Singer stay in Portuguese. Computer exercises are solved in Python (numpy, pandas, statsmodels; no scikit-learn): the answer key shows the code that matters and the key output, and `Listas/L02-book-exercises-check.py` reproduces and asserts every number printed in it. Data: `Listas/data/` (provenance in `Listas/data/README.md`) and the `wooldridge` Python package. Show every step.

Block 1 — *ISLP*, James et al. (2023)

(Ch. 2, 5, 7, 4 • 12 problems)

Question 1. Flexible versus inflexible methods (ISLP 2.1).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 2.1 (Section 2.4, Conceptual), p. 63. Reproduced verbatim.

For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.

(a) The sample size n is extremely large, and the number of predictors p is small.

(b) The number of predictors p is extremely large, and the number of observations n is small.

(c) The relationship between the predictors and response is highly non-linear.

(d) The variance of the error terms, i.e. $\sigma^2 = \mathrm{Var}(\epsilon)$, is extremely high.

Notation.

- $Y = f(X) + \epsilon$: response, f the true regression function, ϵ an error with $E(\epsilon) = 0$ and $\mathrm{Var}(\epsilon) = \sigma^2 > 0$.
- $n \geq 1$: number of training observations; $p \geq 1$: number of predictors.
- $\hat{f}$: the fitted function; x_0: a test point; y_0 its response.
- d: the number of parameters (degrees of freedom) the method fits, $1 \leq d \leq n$; a *flexible* method has a large d.
- $\mathbf{X}$ $(n \times d)$: design matrix; $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$: hat matrix.

Step — the criterion. The expected test MSE at x_0 splits into three non-negative parts (ISLP eq. 2.7):

$$E\big(y_0 - \hat{f}(x_0)\big)^2 = \mathrm{Var}\big(\hat{f}(x_0)\big) + \big[\mathrm{Bias}\big(\hat{f}(x_0)\big)\big]^2 + \mathrm{Var}(\epsilon). \tag{1.1}$$

A flexible method lowers the squared bias and raises the variance; the comparison is decided by which of the two moves more.

Step — how variance scales. For a least-squares fit with d parameters, $\hat{\mathbf{f}} = \mathbf{H}\mathbf{y}$, so

$$\mathrm{Var}(\hat{\mathbf{f}}) = \mathbf{H}\,\mathrm{Var}(\mathbf{y})\,\mathbf{H}'. \tag{1.2}$$

With $\mathrm{Var}(\mathbf{y}) = \sigma^2\mathbf{I}$ and $\mathbf{H}$ symmetric and idempotent, (1.2) becomes

$$\mathrm{Var}(\hat{\mathbf{f}}) = \sigma^2\mathbf{H}. \tag{1.3}$$

Averaging the diagonal of (1.3) over the n training points and using $\mathrm{tr}(\mathbf{H}) = d$:

$$\frac{1}{n}\sum_{i=1}^{n}\mathrm{Var}\big(\hat{f}(x_i)\big) = \frac{\sigma^2\,\mathrm{tr}(\mathbf{H})}{n} = \frac{\sigma^2 d}{n}. \tag{1.4}$$

The variance term of (1.1) therefore grows with d and σ^2 and falls with n.

Step — (a) n large, p small: flexible is better. In (1.4) the ratio d/n stays small even for a large d, so the variance penalty of flexibility is small, while the bias reduction in (1.1) is kept. A flexible method fits f more closely without over-fitting.

Step — (b) p large, n small: flexible is worse. A flexible method needs d of the order of p or more; with n small, d/n in (1.4) approaches 1 and the average variance approaches σ^2: the fit chases noise. An inflexible method (small d) keeps the variance term low; its extra bias costs less than the variance it saves.

Step — (c) highly non-linear f: flexible is better. An inflexible (e.g. linear) method has a squared bias in (1.1) that does not shrink as n grows, because it cannot represent f. A flexible method removes most of that bias; the bias term dominates.

Step — (d) σ^2 very large: flexible is worse. By (1.4) the variance term is proportional to σ^2, so a large σ^2 multiplies the cost of every extra parameter. A flexible method fits the noise; an inflexible one averages it out. (The irreducible term $\mathrm{Var}(\epsilon)$ is the same for both and does not decide.)

Intuition. Flexibility buys lower bias at the price of variance. It pays when data are plentiful relative to the parameters (a) or the truth is far from simple (c); it does not pay when data are scarce relative to the parameters (b) or the signal is buried in noise (d).

Ref: ISLP §2.2.2, "The Bias-Variance Trade-Off", eq. (2.7), pp. 31–34; MS §1.6.1 (pp. 10–14).

Grading rubric (10 points).

- Each of (a)–(d): correct direction (1 pt) and a justification that names the bias or the variance term that dominates (1.5 pt).
- Answering (d) by citing the irreducible error alone, without the variance term: max. 1 pt for (d).

Question 2. Sketch of the bias–variance decomposition (ISLP 2.3).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 2.3 (Section 2.4, Conceptual), p. 63. Reproduced verbatim.

We now revisit the bias-variance decomposition.

(a) Provide a sketch of typical (squared) bias, variance, training error, test error, and Bayes (or irreducible) error curves, on a single plot, as we go from less flexible statistical learning methods towards more flexible approaches. The x-axis should represent the amount of flexibility in the method, and the y-axis should represent the values for each curve. There should be five curves. Make sure to label each one.

(b) Explain why each of the five curves has the shape displayed in part (a).

Notation.

- Flexibility: the effective number of parameters d of the method ($d \geq 1$).
- $\mathrm{Bias}^2(d)$, $\mathrm{Var}(d)$: squared bias and variance of $\hat{f}(x_0)$, averaged over test points.
- $\sigma^2 = \mathrm{Var}(\epsilon) > 0$: irreducible (Bayes) error.
- $\mathrm{MSE}_{\mathrm{tr}}(d)$, $\mathrm{MSE}_{\mathrm{te}}(d)$: expected training and test mean squared errors.

Step — (a) the sketch. The test error is the sum of three curves (ISLP eq. 2.7):

$$\mathrm{MSE}_{\mathrm{te}}(d) = \mathrm{Bias}^2(d) + \mathrm{Var}(d) + \sigma^2. \tag{2.1}$$

Figure 1 draws the five curves. The shapes are illustrative (they are not data): $\mathrm{Bias}^2 = 3e^{-0.45d}$, $\mathrm{Var} = 0.02e^{0.38d}$, $\sigma^2 = 1$, the test curve from (2.1), and a training curve $3.9e^{-0.3d}$.

Step — (b) squared bias. Decreasing: a more flexible class of functions contains (or gets closer to) the true f, so the systematic error of the fit falls, quickly at first and then slowly once f is captured.

Step — (b) variance. Increasing: by (1.4) of Question 1, the average variance of a least-squares fit is $\sigma^2 d/n$, rising with d; a flexible fit changes a lot when one training point changes.

Step — (b) irreducible error. Flat at σ^2: it is the variance of ϵ, which no choice of $\hat{f}$ can predict. It is a floor for the test error in (2.1).

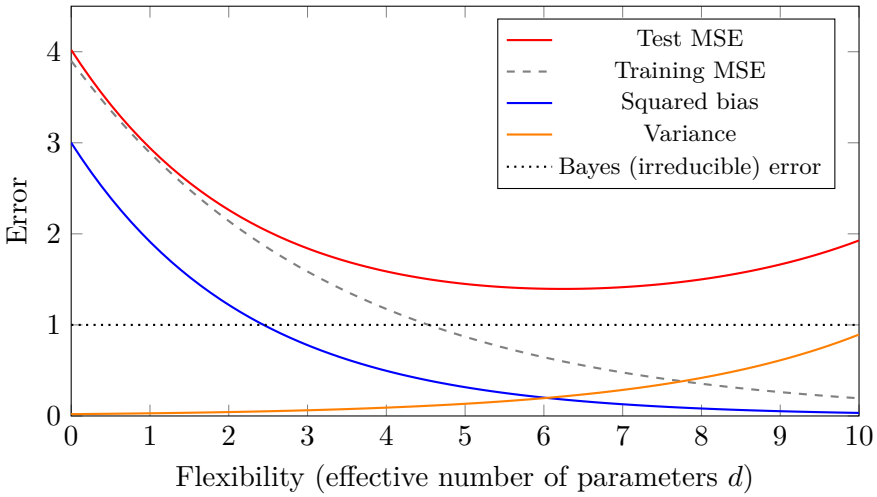


Figure 1: ISLP 2.3(a): the five curves as flexibility grows (illustrative functional forms).

Step — (b) test error. U-shaped: it is the vertical sum (2.1). At low flexibility the falling bias dominates, at high flexibility the rising variance dominates, so the minimum is at an intermediate d, and the curve never goes below σ^2.

Step — (b) training error. Monotonically decreasing, and below the test error: the fit is chosen to make it small, and each extra parameter can only lower it. For a correctly specified least-squares fit with d parameters,

$$E\big[\mathrm{MSE}_{\mathrm{tr}}\big] = \sigma^2 \, \frac{n-d}{n}, \tag{2.2}$$

which falls below σ^2 as d grows and reaches 0 at $d = n$ (interpolation). The gap between test and training error is the optimism that cross-validation corrects.

Intuition. Training error rewards fitting the sample; test error rewards fitting the population. Because the noise in the sample is fitted as if it were signal, the two curves separate as flexibility grows.

Ref: ISLP §2.2.1–2.2.2, Figures 2.9–2.12, pp. 28–34.

Grading rubric (10 points).

- Five labelled curves on one plot, axis labels: 3 pts (1 pt lost per missing or unlabelled curve, at most 3).
- Correct shapes (U-shaped test, decreasing training below test, flat Bayes, decreasing bias, increasing variance): 2 pts.
- Explanation of each shape: 1 pt each (5 pts).
- Training error drawn above the test error or test error below the Bayes line: max. 5 pts in the question.

Question 3. k-fold cross-validation (ISLP 5.3).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 5.3 (Section 5.4, Conceptual), p. 225. Reproduced verbatim.

We now review k-fold cross-validation.

(a) Explain how k-fold cross-validation is implemented.

(b) What are the advantages and disadvantages of k-fold cross-validation relative to:

i. The validation set approach?

ii. LOOCV?

Notation.

- n: number of observations; $k \in \{2, \ldots, n\}$: number of folds; $C_1, \ldots, C_k$: the folds, a random partition of $\{1, \ldots, n\}$ with sizes $n_j \approx n/k$.
- $\hat{f}^{(-j)}$: the model fitted without fold j; MSE_j: its error on fold j.
- $\text{CV}_{(k)}$: the k-fold estimate of the test MSE; $\rho \in [0, 1]$: the correlation between two fold errors; $\tau^2 > 0$: the variance of one fold error.

Step — (a) the algorithm. (1) Shuffle the indices and cut them into k folds of (nearly) equal size. (2) For $j = 1, \ldots, k$: fit the method on the other $k - 1$ folds and compute

$$\text{MSE}_j = \frac{1}{n_j} \sum_{i \in C_j} \left(y_i - \hat{f}^{(-j)}(x_i)\right)^2. \tag{3.1}$$

(3) Average (ISLP eq. 5.3):

$$\text{CV}_{(k)} = \frac{1}{k} \sum_{j=1}^{k} \text{MSE}_j. \tag{3.2}$$

For classification, replace the squared error in (3.1) by the indicator $I(y_i \neq \hat{y}_i)$. To choose a tuning parameter (degree, K), repeat with the same folds for each candidate and pick the smallest $\text{CV}_{(k)}$. Each model is trained on

$$n_{\text{train}} = n \, \frac{k-1}{k} \tag{3.3}$$

observations.

Step — (b)(i) versus the validation set approach. *Advantages.* (1) Lower variance: the validation-set estimate depends on one random split, while (3.2) averages k errors and every observation is used once for validation. (2) Lower bias: by (3.3) with $k = 10$ and $n = 100$, $n_{\text{train}} = 100 \cdot \frac{9}{10} = 90$, against $n/2 = 50$ for a half split; a model trained on fewer points is worse, so the validation set approach overstates the test error. *Disadvantage.* k fits instead of one (computational cost).

Step — (b)(ii) versus LOOCV. LOOCV is (3.2) with $k = n$. *Advantages of k-fold.* (1) Cost: k fits instead of n (except for least squares, where the shortcut ISLP eq. 5.2 gives LOOCV from one fit). (2) Lower variance: the n LOOCV fits share $n - 2$ observations, so their errors are highly correlated. With equicorrelated fold errors,

$$\text{Var}\big(\text{CV}_{(k)}\big) = \frac{\tau^2}{k}\big(1 + (k-1)\rho\big), \tag{3.4}$$

and as $\rho \to 1$ (3.4) tends to τ^2: averaging nearly identical errors does not reduce variance. *Disadvantage of k-fold.* More bias: by (3.3) it trains on $n(k-1)/k$ rather than $n - 1$ points (e.g. 90 versus 99 for $n = 100$), so it slightly overstates the test error. $k = 5$ or 10 is the usual compromise.

Intuition. The three methods trade the bias of training on fewer points against the variance of the estimate. The validation set is cheap but noisy and pessimistic; LOOCV is almost unbiased but its n near-identical fits make it noisy; k-fold sits in between.

Ref: ISLP §5.1.1–5.1.4, pp. 202–209; MS §8.6, Nota 1 (pp. 285–287); AME §1.5.1 (pp. 14–21).

Grading rubric (10 points).

- (a) Partition, fit on $k - 1$ folds, error on the held-out fold, average (3.2): 4 pts.
- (b)(i) Variance and bias advantages plus the cost: 3 pts.
- (b)(ii) Cost, variance (correlated fits) and bias: 3 pts.
- Claiming LOOCV has *higher* bias than k-fold: max. 1 pt in (b)(ii).

Question 4. Validation-set error of a logistic regression on `Default` (ISLP 5.5).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 5.5 (Section 5.4, Applied), pp. 225–226. Reproduced verbatim.

In Chapter 4, we used logistic regression to predict the probability of `default` using `income` and `balance` on the `Default` data set. We will now estimate the test error of this logistic regression model using the validation set approach. Do not forget to set a random seed before beginning your analysis.

(a) Fit a logistic regression model that uses `income` and `balance` to predict `default`.

(b) Using the validation set approach, estimate the test error of this model. In order to do this, you must perform the following steps:

i. Split the sample set into a training set and a validation set.

ii. Fit a multiple logistic regression model using only the training observations.

iii. Obtain a prediction of default status for each individual in the validation set by computing the posterior probability of default for that individual, and classifying the individual to the `default` category if the posterior probability is greater than 0.5.

iv. Compute the validation set error, which is the fraction of the observations in the validation set that are misclassified.

(c) Repeat the process in (b) three times, using three different splits of the observations into a training set and a validation set. Comment on the results obtained.

(d) Now consider a logistic regression model that predicts the probability of `default` using `income`, `balance`, and a dummy variable for `student`. Estimate the test error for this model using the validation set approach. Comment on whether or not including a dummy variable for `student` leads to a reduction in the test error rate.

Notation.

- $Y_i \in \{0,1\}$: default indicator ($1 =$ `Yes`); X_1: `income` (USD); X_2: `balance` (USD); $X_3 \in \{0,1\}$: `student`; $n = 10{,}000$.
- $p(x) = \Pr(Y = 1 \mid X = x) \in (0,1)$; $\beta_0, \beta_1, \beta_2, \beta_3$: logistic coefficients.
- T, V: training and validation index sets, $|T| = |V| = 5000$; s: the seed of `numpy`'s `default_rng(s)`, whose `permutation` defines the split.
- Err_V: validation error rate.

Step — model and rule. The multiple logistic model (ISLP eq. 4.7) is

$$p(x) = \frac{e^{\beta_0+\beta_1 x_1+\beta_2 x_2}}{1 + e^{\beta_0+\beta_1 x_1+\beta_2 x_2}}, \tag{4.1}$$

the classification rule is

$$\hat{y} = 1 \iff \hat{p}(x) > 0.5, \tag{4.2}$$

and the validation error is

$$\mathrm{Err}_V = \frac{1}{|V|} \sum_{i \in V} I(y_i \neq \hat{y}_i). \tag{4.3}$$

Step — (a) full-sample fit. Maximum likelihood on all 10,000 observations gives $\hat{\beta}_0 = -11.5405$, $\hat{\beta}_1 = 2.0809 \times 10^{-5}$ (s.e. 4.985×10^{-6}), $\hat{\beta}_2 = 5.6471 \times 10^{-3}$ (s.e. 2.274×10^{-4}). Both slopes are significant ($z = 4.17$ and 24.84). In odds terms, an extra 100 dollars of balance multiplies the odds of default by

$$e^{100\hat{\beta}_2} = e^{100\,(0.0056471)} = e^{0.56471} = 1.7589. \tag{4.4}$$

Step — (b) one split ($s = 1$). Fit (4.1) on T, classify V with (4.2). Confusion matrix on V (rows: predicted; columns: true):

	true No	true Yes
predicted No	4813	110
predicted Yes	22	55

The misclassified are the off-diagonal cells, $110 + 22 = 132$, so by (4.3)

$$\text{Err}_V = \frac{132}{5000} = 0.0264. \tag{4.5}$$

Step — (c) three more splits. Seeds $s = 2, 3, 4$ give $\text{Err}_V = 0.0256$, 0.0296, 0.0264. Averaging with (4.5):

$$\overline{\text{Err}} = \frac{0.0264 + 0.0256 + 0.0296 + 0.0264}{4} = \frac{0.1080}{4} = 0.0270. \tag{4.6}$$

The estimate moves by $0.0296 - 0.0256 = 0.0040$ across splits: the validation set approach is noisy because it depends on which 5000 people land in V. All values are below the error of the null classifier that always predicts No, 0.0333 (the default rate).

Step — (d) adding `student`. With X_3 in (4.1) and the same four splits: 0.0270, 0.0264, 0.0294, 0.0264, so

$$\overline{\text{Err}}_{\text{student}} = \frac{0.0270 + 0.0264 + 0.0294 + 0.0264}{4} = \frac{0.1092}{4} = 0.0273. \tag{4.7}$$

Comparing (4.7) with (4.6), the dummy does not reduce the test error: the difference, 0.0003, is an order of magnitude smaller than the split-to-split range 0.0040. Given `balance`, `student` carries little extra information for classification.

```
dflt = pd.read_csv("data/Default.csv"); y = (dflt.default == "Yes").astype(int).to_numpy()
X = dflt[["income", "balance"]].to_numpy()
for s in (1, 2, 3, 4):
    idx = np.random.default_rng(s).permutation(len(y)); tr, va = idx[:5000], idx[5000:]
    res = sm.Logit(y[tr], sm.add_constant(X[tr])).fit(disp=0)
    err = ((res.predict(sm.add_constant(X[va])) > 0.5) != y[va]).mean()
```

Intuition. The error rate is dominated by the rare defaulters; whether 10 more of them fall in the validation half moves the estimate as much as any change of model. Comparing models on a single split is fragile; that is why ISLP moves on to LOOCV and k-fold CV.

Ref: ISLP §5.1.1 (validation set approach, pp. 202–204); §4.3.3–4.3.4 (multiple logistic regression, `Default`, pp. 141–144).

Grading rubric (10 points).

- (a) Fit and report the coefficients: 2 pts.
- (b) Split, train-only fit, threshold 0.5, misclassification rate: 3 pts.
- (c) Three other splits and a comment on split variability: 2 pts.
- (d) Model with `student`, comparison relative to the split noise: 3 pts.
- Error computed on the training set instead of V: max. 4 pts in the question.

Question 5. LOOCV for the polynomial degree on simulated data (ISLP 5.8).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 5.8 (Section 5.4, Applied), p. 227. Reproduced verbatim.

We will now perform cross-validation on a simulated data set.

(a) Generate a simulated data set as follows:

```
rng = np.random.default_rng(1)
x = rng.normal(size=100)
y = x - 2 * x**2 + rng.normal(size=100)
```

In this data set, what is n and what is p? Write out the model used to generate the data in equation form.

(b) Create a scatterplot of X against Y. Comment on what you find.

(c) Set a random seed, and then compute the LOOCV errors that result from fitting the following four models using least squares:

i. $Y = \beta_0 + \beta_1 X + \epsilon$

ii. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$

iii. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$

iv. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \epsilon$.

Note you may find it helpful to use the `data.frame()` function to create a single data set containing both X and Y.

(d) Repeat (c) using another random seed, and report your results. Are your results the same as what you got in (c)? Why?

(e) Which of the models in (c) had the smallest LOOCV error? Is this what you expected? Explain your answer.

(f) Comment on the statistical significance of the coefficient estimates that results from fitting each of the models in (c) using least squares. Do these results agree with the conclusions drawn based on the cross-validation results?

Notation.

- $X_i \sim N(0,1)$, $\epsilon_i \sim N(0,1)$ independent, $i = 1, \ldots, n$; Y_i: response.
- $m \in \{1,2,3,4\}$: polynomial degree of model i–iv; $\mathbf{X}_m$: the $n \times (m+1)$ matrix with columns $1, X, \ldots, X^m$.
- $\mathbf{H}_m = \mathbf{X}_m(\mathbf{X}_m'\mathbf{X}_m)^{-1}\mathbf{X}_m'$; h_i: its i-th diagonal element (leverage), $1/n \le h_i \le 1$.
- $\hat{y}_i$: full-sample fitted value; $\mathrm{CV}_{(n)}$: LOOCV estimate of the test MSE.

Step — (a) n, p and the model. $n = 100$ observations and $p = 1$ predictor (X; the model uses X and X^2, but both are functions of one variable). The data-generating process is

$$Y = X - 2X^2 + \epsilon, \qquad X \sim N(0,1),\ \epsilon \sim N(0,1), \tag{5.1}$$

i.e. model ii with $\beta_0 = 0$, $\beta_1 = 1$, $\beta_2 = -2$ and $\sigma^2 = 1$.

Step — (b) the scatterplot. Figure 2 shows an inverted parabola. From (5.1), $E(Y \mid X = x) = x - 2x^2$; setting its derivative to zero,

$$1 - 4x = 0 \ \Rightarrow\ x = 0.25, \tag{5.2}$$

and substituting (5.2) into (5.1), the peak is $0.25 - 2(0.25)^2 = 0.25 - 0.125 = 0.125$. The sample y ranges from -14.864 to 2.097: the quadratic term makes large $|x|$ produce very negative y, so a straight line cannot fit.

Step — (c) LOOCV errors. For least squares, LOOCV has the closed form (ISLP eq. 5.2)

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2, \tag{5.3}$$

with h_i the diagonal of $\mathbf{H}_m$. Evaluating (5.3) (and checking it against 100 explicit leave-one-out fits, which agree to 10^{-10}):

model	i ($m=1$)	ii ($m=2$)	iii ($m=3$)	iv ($m=4$)
$\mathrm{CV}_{(n)}$	6.6330	1.1229	1.3018	1.3324

Step — (d) another seed. The results are identical. LOOCV leaves out each observation exactly once, so (5.3) is a deterministic function of the data; there is no random split for the seed to change (the data themselves were fixed in (a)).

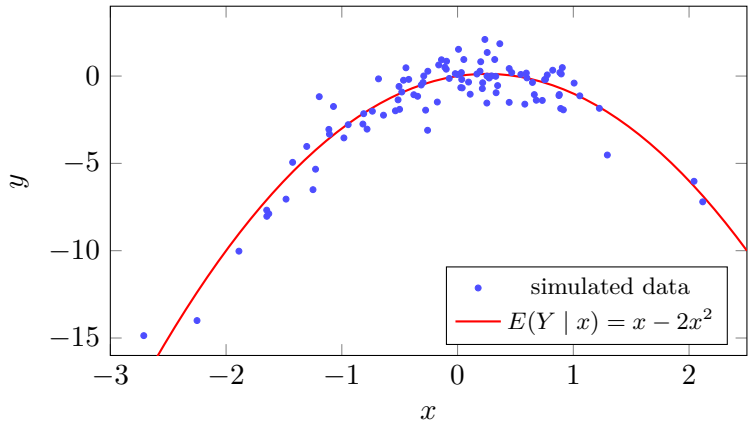


Figure 2: ISLP 5.8(b): the 100 simulated points and the true regression function.

Step — (e) the smallest error. Model ii, $\mathrm{CV}_{(n)} = 1.1229$. This is expected: by (5.1) the truth is quadratic, so model i has a large bias (6.6330), and models iii–iv add terms whose true coefficients are zero, which only adds variance. For a correctly specified model with $m+1$ coefficients the expected test MSE is about $\sigma^2(1 + \frac{m+1}{n}) = 1 \cdot (1 + \frac{3}{100}) = 1.03$, close to 1.1229.

Step — (f) significance. OLS t-statistics (and p-values) on the full sample:

model	$\hat\beta_0$	$\hat\beta_1$	$\hat\beta_2$	$\hat\beta_3$	$\hat\beta_4$
i	−5.94 (0.000)	6.75 (0.000)			
ii	−0.61 (0.543)	7.65 (0.000)	−22.07 (0.000)		
iii	−0.48 (0.635)	5.95 (0.000)	−20.67 (0.000)	−1.07 (0.287)	
iv	0.74 (0.460)	4.42 (0.000)	−11.34 (0.000)	0.47 (0.642)	2.31 (0.023)

The linear and quadratic terms are always highly significant; the cubic term never is. The quartic term has $p = 0.023$, which by itself would suggest model iv, but LOOCV rejects it ($1.3324 > 1.1229$). This is a false positive of the kind expected when many coefficients are tested at 5%, helped by the extreme x values (the largest leverage is $h_i = 0.836$ in model iv), which a fourth power fits closely. Overall the tests and CV agree on model ii; where they disagree, CV measures what matters for prediction.

```
for m in (1, 2, 3, 4):
    B = np.column_stack([x**j for j in range(m + 1)])
    H = B @ np.linalg.solve(B.T @ B, B.T)
    cv = np.mean(((y - H @ y) / (1 - np.diag(H)))**2)       # (5.2)
    fit = sm.OLS(y, B).fit()                                  # t-stats, p-values for (f)
```

Intuition. LOOCV estimates out-of-sample error directly, so it penalizes the extra variance of superfluous terms automatically, while each t-test asks only whether one coefficient is zero.

Ref: ISLP §5.1.2 (LOOCV and eq. 5.2, pp. 204–206); §3.3.2 (non-linear relationships, p. 98).

Grading rubric (10 points).

- (a) $n = 100$, $p = 1$, model (5.1): 1 pt. (b) plot and comment: 1 pt.
- (c) four LOOCV errors: 3 pts. (d) identical, with the reason: 1.5 pts.
- (e) model ii and why: 1.5 pts. (f) significance and its relation to CV: 2 pts.
- Saying the results change with the seed in (d): max. 0.5 pt in (d).

Question 6. CV-chosen polynomial degree and step functions for `Wage` (ISLP 7.6).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 7.6 (Section 7.9, Applied), p. 327. Reproduced verbatim.

In this exercise, you will further analyze the `Wage` data set considered throughout this chapter.

(a) Perform polynomial regression to predict `wage` using `age`. Use cross-validation to select the optimal degree d for the polynomial. What degree was chosen, and how does this compare to the results of hypothesis testing using ANOVA? Make a plot of the resulting polynomial fit to the data.

(b) Fit a step function to predict `wage` using `age`, and perform cross-validation to choose the optimal number of cuts. Make a plot of the fit obtained.

Notation.

- y_i: `wage` (thousands of USD); x_i: `age` (18–80 years); $n = 3000$.
- $d \in \{1, \dots, 10\}$: polynomial degree; RSS_d: residual sum of squares of the degree-d fit; $\mathrm{CV}_{(10)}(d)$: 10-fold CV estimate of the test MSE.
- $c \in \{2, \dots, 10\}$: number of intervals of the step function ($c - 1$ cut points), equal width over $[18, 80]$; $C_1(x), \dots, C_{c-1}(x)$: interval dummies (the first interval is the base).
- Folds: `numpy.random.default_rng(1).permutation(n) % 10`, the same for every model.

Step — model and criterion. The polynomial regression (ISLP eq. 7.1) is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \epsilon_i, \tag{6.1}$$

fitted by least squares (with x standardized before taking powers, which changes no fitted value but avoids ill-conditioning). For each d the 10-fold CV error is (3.2) of Question 3 with $k = 10$.

Step — (a) CV choice. The 10-fold CV errors are:

d	1	2	3	4	5	6	7	8	9	10
$\mathrm{CV}_{(10)}$	1675.48	1599.65	1594.81	1593.32	1593.23	1592.32	1592.06	1592.70	1590.97	1592.11

The minimum is at $d = 9$ (repeating the CV with fold seeds $2, \dots, 5$, the minimum is at $d = 6$ for seed 2 and at $d = 9$ for seeds 3, 4 and 5). But the curve is flat beyond $d = 3$–4: the gain from $d = 4$ to $d = 9$ is

$$1593.32 - 1590.97 = 2.35, \qquad \frac{2.35}{1593.32} = 0.0015, \tag{6.2}$$

a 0.15% reduction, far inside the sampling noise of a CV estimate. The big step is from $d = 1$ to $d = 2$ (75.83), with a small further gain to $d = 3$.

Step — (a) ANOVA. Nested F-tests compare degree $d - 1$ with d, using the residual variance of the largest model ($d = 5$), as `anova_lm` does:

$$F_d = \frac{\mathrm{RSS}_{d-1} - \mathrm{RSS}_d}{\mathrm{RSS}_5/(n-6)}, \qquad F_d \sim F_{1,\,n-6} \text{ under } H_0 : \beta_d = 0. \tag{6.3}$$

The denominator of (6.3) is

$$\frac{\mathrm{RSS}_5}{n-6} = \frac{4770321.69}{3000-6} = \frac{4770321.69}{2994} = 1593.29. \tag{6.4}$$

Substituting (6.4) into (6.3), e.g. for $d = 3$:

$$F_3 = \frac{4793430.09 - 4777674.40}{1593.29} = \frac{15755.69}{1593.29} = 9.889. \tag{6.5}$$

All four comparisons:

comparison	$\text{RSS}_{d-1} - \text{RSS}_d$	F_d	p-value
$d = 1 \to 2$	$5022216.10 - 4793430.09 = 228786.01$	143.59	< 0.001
$d = 2 \to 3$	$4793430.09 - 4777674.40 = 15755.69$	9.889	0.0017
$d = 3 \to 4$	$4777674.40 - 4771604.25 = 6070.15$	3.810	0.0510
$d = 4 \to 5$	$4771604.25 - 4770321.69 = 1282.56$	0.805	0.3697

ANOVA keeps the cubic (and the quartic at the 10% level only) and rejects $d = 5$. CV and ANOVA agree on what matters: degree 1 and 2 are clearly too rigid and nothing beyond $d \approx 4$ buys a material improvement; CV's literal minimizer $d = 9$ is within noise of $d = 3$–4. A parsimonious choice is $d = 4$, plotted in Figure 3:

$$\widehat{\texttt{wage}} = -184.154 + 21.2455\,x - 0.563859\,x^2 + 0.00681069\,x^3 - 0.0000320383\,x^4, \tag{6.6}$$

which peaks at age 48.35.

Step — (b) step function. The step-function model (ISLP eq. 7.5) is

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_{c-1} C_{c-1}(x_i) + \epsilon_i, \tag{6.7}$$

whose fit is the mean wage within each interval. With interval edges fixed on the full range $[18, 80]$ and the same folds:

c	2	3	4	5	6	7	8	9	10
$\text{CV}_{(10)}$	1733.16	1681.80	1634.27	1629.82	1621.38	1609.87	1599.57	1607.75	1602.75

CV selects $c = 8$ intervals (7 cuts), with edges $18, 25.75, 33.5, 41.25, 49, 56.75, 64.5, 72.25, 80$ and fitted levels $76.28, 102.12, 116.51, 119.78, 116.42, 120.38, 105.23, 91.51$. Even the best step function (1599.57) does worse than the cubic (1594.81): the steps cannot follow the smooth rise of wages in the 20s.

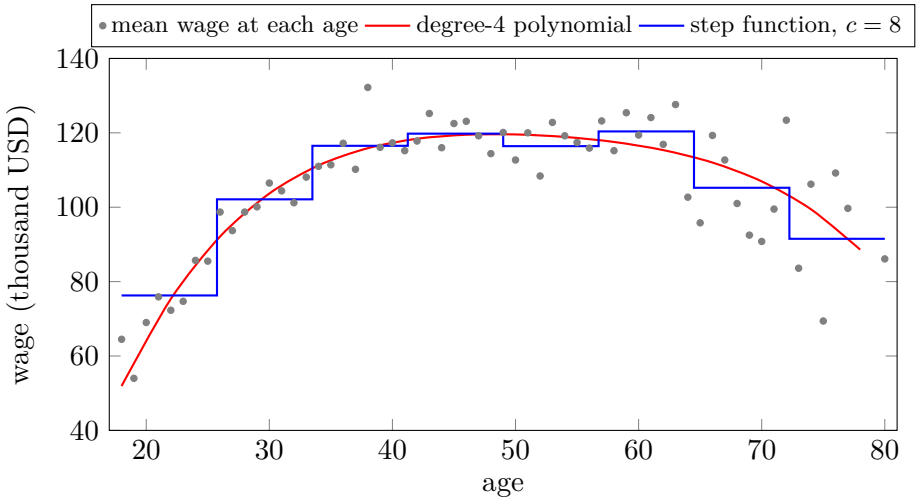


Figure 3: ISLP 7.6: polynomial and step-function fits (the 3000 points are summarized by the mean wage at each age).

```
wage = pd.read_csv("data/Wage.csv"); age, wg = wage.age.to_numpy(float), wage.wage.to_numpy()
basis = lambda a, d: np.column_stack([((a - age.mean()) / age.std())**j for j in range(d + 1)])
cv = {d: cv_mse(lambda a: basis(a, d), age, wg, k=10, seed=1) for d in range(1, 11)}
edges = np.linspace(18, 80, c + 1)                          # step function with c intervals
step = lambda a: np.column_stack([np.ones(len(a))] + [((a > edges[j]) & (a <= edges[j+1]))
                                  .astype(float) for j in range(1, c)])
```

Intuition. Wages rise steeply until the mid-30s, flatten, and fall after 60. Three to four polynomial terms capture this shape; more terms only wiggle around it, which CV can barely distinguish from no change. Step functions pay a bias price for being flat inside each bin.

Ref: ISLP §7.1–7.2 (pp. 290–293), §7.8.1 lab (ANOVA for the polynomial degree, p. 310); §5.1.3 (k-fold CV, p. 206).

Grading rubric (10 points).

- (a) CV over degrees with the table and the selected degree: 3 pts; ANOVA F-tests and comparison: 2 pts; plot: 1 pt.
- (b) Step function, CV over the number of cuts, selected value: 3 pts; plot: 1 pt.
- Choosing the degree by training RSS (always the largest d): max. 2 pts in (a).

Question 7. Polynomials and regression splines for `nox` on `dis` (ISLP 7.9).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 7.9 (Section 7.9, Applied), p. 327. Reproduced verbatim.

This question uses the variables `dis` (the weighted mean of distances to five Boston employment centers) and `nox` (nitrogen oxides concentration in parts per 10 million) from the `Boston` data. We will treat `dis` as the predictor and `nox` as the response.

(a) Use the `poly()` function from the `ISLP.models` module to fit a cubic polynomial regression to predict `nox` using `dis`. Report the regression output, and plot the resulting data and polynomial fits.

(b) Plot the polynomial fits for a range of different polynomial degrees (say, from 1 to 10), and report the associated residual sum of squares.

(c) Perform cross-validation or another approach to select the optimal degree for the polynomial, and explain your results.

(d) Use the `bs()` function from the `ISLP.models` module to fit a regression spline to predict `nox` using `dis`. Report the output for the fit using four degrees of freedom. How did you choose the knots? Plot the resulting fit.

(e) Now fit a regression spline for a range of degrees of freedom, and plot the resulting fits and report the resulting RSS. Describe the results obtained.

(f) Perform cross-validation or another approach in order to select the best degrees of freedom for a regression spline on this data. Describe your results.

Notation.

- y_i: `nox` (parts per 10 million); x_i: `dis` (from 1.13 to 12.13); $n = 506$.
- d: polynomial degree; $\text{RSS} = \sum_i (y_i - \hat{y}_i)^2$; $\text{CV}_{(10)}$: 10-fold CV error, folds from `default_rng(1)`, identical for every model.
- $\xi_1 < \cdots < \xi_K$: interior knots of a cubic spline; df: number of spline basis columns excluding the intercept; $B_j(x)$: cubic B-spline basis functions.

Step — (a) cubic fit. Least squares on $1, x, x^2, x^3$ (`poly()` uses an orthogonal basis by default; the fitted values, R^2 and RSS are the same, and the raw coefficients are easier to read):

$$\widehat{\texttt{nox}} = \underset{(0.0207)}{0.9341} - \underset{(0.0147)}{0.1821}\, x + \underset{(0.00293)}{0.02193}\, x^2 - \underset{(0.000173)}{0.000885}\, x^3, \tag{7.1}$$

standard errors in parentheses, every p-value < 0.001, $R^2 = 0.7148$, $\text{RSS} = 1.9341$. Figure 4 plots the data and (7.1): `nox` falls steeply as distance grows up to about 6 and then levels off; the cubic bends down again at the right edge, where data are sparse.

Step — (b) degrees 1 to 10. Residual sums of squares:

d	1	2	3	4	5	6	7	8	9	10
RSS	2.7686	2.0353	1.9341	1.9330	1.9153	1.8783	1.8495	1.8356	1.8333	1.8322

RSS falls monotonically because the models are nested (each adds one column): training error cannot rise with flexibility (Question 2). The drop is large up to $d = 3$ and small afterwards.

Step — (c) CV for the degree. 10-fold CV errors (multiplied by 10^3):

d	1	2	3	4	5	6	7	8	9	10
$10^3\,\mathrm{CV}_{(10)}$	5.526	4.065	3.852	3.867	4.184	5.811	8.736	8.281	12.092	3.987

CV selects $d = 3$. Unlike RSS, the CV error rises again for $d \geq 5$ and is erratic: a high-degree polynomial fitted without the few observations at large `dis` extrapolates wildly at those points when they are held out. The cubic balances bias and variance.

Step — (d) spline with 4 df. A cubic spline with K interior knots has, excluding the intercept,

$$\mathrm{df} = 3 + K \tag{7.2}$$

basis functions. With df $= 4$, (7.2) gives $K = 4 - 3 = 1$ knot. Knots are placed at uniform quantiles of x (the default of `bs()`), so the single knot is the median, $\xi_1 = 3.2074$. The model is

$$y_i = \beta_0 + \sum_{j=1}^{4} \beta_j B_j(x_i) + \epsilon_i, \tag{7.3}$$

with estimates (standard errors) $\hat{\beta}_0 = 0.7345\,(0.0146)$, $\hat{\beta}_1 = -0.0581\,(0.0219)$, $\hat{\beta}_2 = -0.4636\,(0.0237)$, $\hat{\beta}_3 = -0.1998\,(0.0431)$, $\hat{\beta}_4 = -0.3888\,(0.0455)$; RSS $= 1.9228$, $R^2 = 0.7164$. The B-spline coefficients have no direct meaning (each weights a local bump); the fit (Figure 4) is close to the cubic but bends less at the right edge.

Step — (e) a range of df. RSS of cubic regression splines:

df	3	4	5	6	7	8	9	10
RSS	1.9341	1.9228	1.8402	1.8340	1.8299	1.8170	1.8257	1.7925

With df $= 3$, (7.2) gives $K = 0$: the spline is the global cubic, and its RSS, 1.9341, equals the cubic RSS of (b), a useful check. RSS falls with df but not monotonically ($1.8257 > 1.8170$ at df $= 9$), because changing df moves all the quantile knots, so the models are not nested.

Step — (f) CV for the df. 10-fold CV errors (multiplied by 10^3):

df	3	4	5	6	7	8	9	10
$10^3\,\mathrm{CV}_{(10)}$	3.852	3.875	3.688	3.687	3.700	3.690	3.715	3.695

The minimum is at df $= 6$ (0.003687), but df $= 5$ to 10 are all within 0.00003 of it; any of them improves on the global cubic (0.003852), and df $= 5$ is the parsimonious choice. Splines do not suffer the erratic CV of high-degree polynomials because each basis function is local.

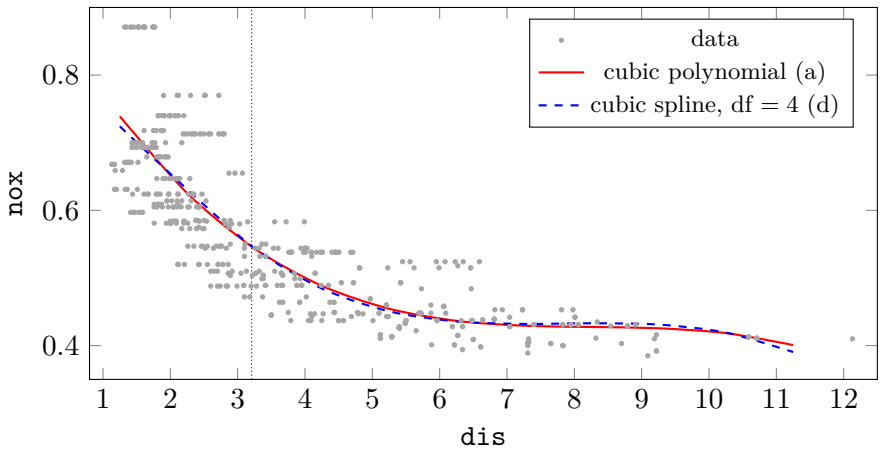


Figure 4: ISLP 7.9: data, cubic polynomial and the 4-df cubic spline (dotted line: the knot at the median of `dis`).

```
def bs_basis(x, df, xall):                          # cubic B-splines, knots at quantiles of xall
    inner = np.quantile(xall, np.linspace(0, 1, df - 1)[1:-1])
    t = np.r_[[xall.min()] * 4, inner, [xall.max()] * 4]
    B = BSpline.design_matrix(np.clip(x, xall.min(), xall.max()), t, 3).toarray()
    return np.column_stack([np.ones(len(x)), B[:, 1:]])      # drop one column: df columns
fit = sm.OLS(nox, bs_basis(dis, 4, dis)).fit()
```

Intuition. Pollution falls with distance from the employment centers and flattens in the suburbs. A global polynomial must use the same coefficients everywhere, so getting the curvature right near the center forces strange behaviour at the far end; a spline adapts locally around its knots.

Ref: ISLP §7.1 (polynomial regression, pp. 290–292), §7.4 (regression splines, pp. 294–299), especially §7.4.4 on choosing the number and location of knots.

Grading rubric (10 points).

- (a) Output and plot: 1.5 pts. (b) RSS for 10 degrees and why it falls: 1.5 pts.
- (c) CV choice with explanation: 2 pts. (d) spline with 4 df, knot choice, output, plot: 2 pts.
- (e) RSS over df with comment: 1.5 pts. (f) CV choice of df: 1.5 pts.
- Selecting the degree or df by RSS alone: max. half of the points of (c) or (f).

Question 8. Logistic function and logit are equivalent (ISLP 4.1).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.1 (Section 4.8, Conceptual), p. 193. Reproduced verbatim. Equations (4.2) and (4.3) of the book are restated in the solution.

Using a little bit of algebra, prove that (4.2) is equivalent to (4.3). In other words, the logistic function representation and logit representation for the logistic regression model are equivalent.

Notation.

- $p(X) = \Pr(Y = 1 \mid X) \in (0, 1)$; $\beta_0, \beta_1 \in \mathbb{R}$; $\eta = \beta_0 + \beta_1 X$ (the linear predictor), so $e^\eta > 0$.

Step — the two representations. The logistic function (ISLP eq. 4.2) and the odds form (ISLP eq. 4.3) are

$$p(X) = \frac{e^\eta}{1 + e^\eta}, \tag{8.1}$$

$$\frac{p(X)}{1 - p(X)} = e^\eta. \tag{8.2}$$

Step — (8.1) ⇒ (8.2). From (8.1), the complement is

$$1 - p(X) = 1 - \frac{e^\eta}{1 + e^\eta} = \frac{1 + e^\eta - e^\eta}{1 + e^\eta} = \frac{1}{1 + e^\eta}. \tag{8.3}$$

Dividing (8.1) by (8.3):

$$\frac{p(X)}{1 - p(X)} = \frac{\left(\dfrac{e^\eta}{1 + e^\eta}\right)}{\left(\dfrac{1}{1 + e^\eta}\right)}. \tag{8.4}$$

Cancelling the common factor $1 + e^\eta > 0$ in (8.4) gives e^η, which is (8.2).

Step — (8.2) ⇒ (8.1). Multiplying (8.2) by $1 - p(X) > 0$:

$$p(X) = e^\eta\big(1 - p(X)\big). \tag{8.5}$$

Expanding and collecting the $p(X)$ terms of (8.5) on the left:

$$p(X)\big(1 + e^\eta\big) = e^\eta. \tag{8.6}$$

Dividing (8.6) by $1 + e^{\eta} > 0$ gives (8.1). Taking logs of (8.2) also gives the logit form $\log\{p(X)/[1 - p(X)]\} = \beta_0 + \beta_1 X$ (ISLP eq. 4.4).

Intuition. The logistic function maps the real line onto $(0, 1)$ and the logit is its inverse; they are two ways of writing the same model, one on the probability scale and one on the log-odds scale, where it is linear.

Ref: ISLP §4.3.1, eqs. (4.2)–(4.4), pp. 139–140; MS §8.2, eq. (8.2), p. 264.

Grading rubric (10 points).

- Direction (8.1)⇒(8.2) with $1 - p$ computed: 5 pts.
- Converse direction (or an argument that the map is a bijection): 5 pts.
- Proving one direction only: max. 5 pts.

Question 9. Odds and probabilities (ISLP 4.9).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.9 (Section 4.8, Conceptual), p. 195. Reproduced verbatim.

This problem has to do with *odds*.

(a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?

(b) Suppose that an individual has a 16 % chance of defaulting on her credit card payment. What are the odds that she will default?

Notation.

- $p \in (0, 1)$: probability of default; $o = p/(1 - p) \in (0, \infty)$: odds of default.

Step — the two maps. By definition, and by solving it for p (the steps of Question 8),

$$o = \frac{p}{1 - p}, \qquad p = \frac{o}{1 + o}. \tag{9.1}$$

Step — (a). Substituting $o = 0.37$ into the second form of (9.1):

$$p = \frac{(0.37)}{1 + (0.37)} = \frac{0.37}{1.37} = 0.2701. \tag{9.2}$$

About 27% of such people default.

Step — (b). Substituting $p = 0.16$ into the first form of (9.1):

$$o = \frac{(0.16)}{1 - (0.16)} = \frac{0.16}{0.84} = 0.1905. \tag{9.3}$$

The odds are about 0.19 (roughly 4 to 21 against).

Intuition. Odds and probability are close only when the event is rare; at $p = 0.16$ the odds already exceed p by $0.1905 - 0.16 = 0.0305$, and the gap grows quickly as p rises.

Ref: ISLP §4.3.1, eq. (4.3) and the discussion of odds, p. 140.

Grading rubric (10 points).

- (a) $p = o/(1 + o)$ and 0.2701: 5 pts. (b) $o = p/(1 - p)$ and 0.1905: 5 pts.
- Confusing odds with probability (answering 0.37 or 0.16): 0 pts for that item.

Question 10. Logistic regression versus the softmax coding (ISLP 4.12).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.12 (Section 4.8, Conceptual), p. 196. Reproduced verbatim, including a misprint in item (d) flagged in the note after the statement.

Suppose that you wish to classify an observation $X \in \mathbb{R}$ into `apples` and `oranges`. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \texttt{orange} \mid X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}.$$

Your friend fits a logistic regression model to the same data using the *softmax* formulation in (4.13), and finds that

$$\widehat{\Pr}(Y = \texttt{orange} \mid X = x) = \frac{\exp(\hat{\alpha}_{\texttt{orange0}} + \hat{\alpha}_{\texttt{orange1}} x)}{\exp(\hat{\alpha}_{\texttt{orange0}} + \hat{\alpha}_{\texttt{orange1}} x) + \exp(\hat{\alpha}_{\texttt{apple0}} + \hat{\alpha}_{\texttt{apple1}} x)}.$$

(a) What is the log odds of `orange` versus `apple` in your model?

(b) What is the log odds of `orange` versus `apple` in your friend's model?

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.

(d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{\texttt{orange0}} = 1.2$, $\hat{\alpha}_{\texttt{orange1}} = -2$, $\hat{\alpha}_{\texttt{orange0}} = 3$, $\hat{\alpha}_{\texttt{orange1}} = 0.6$. What are the coefficient estimates in your model?

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

Note (not in the book): in (d) the book prints the second pair as $\hat{\alpha}_{\texttt{orange0}} = 3$, $\hat{\alpha}_{\texttt{orange1}} = 0.6$; it is a misprint for $\hat{\alpha}_{\texttt{apple0}} = 3$, $\hat{\alpha}_{\texttt{apple1}} = 0.6$, and is read that way below.

Notation.
- $p_o(x) = \widehat{\Pr}(Y = \texttt{orange} \mid X = x)$, $p_a(x) = 1 - p_o(x)$.
- $\hat{\beta}_0, \hat{\beta}_1$: your (baseline-coded) coefficients; $\hat{\alpha}_{o0}, \hat{\alpha}_{o1}, \hat{\alpha}_{a0}, \hat{\alpha}_{a1}$: your friend's softmax coefficients.

Step — (a) your log odds. By Question 8, (8.2) in logs:

$$\log \frac{p_o(x)}{p_a(x)} = \hat{\beta}_0 + \hat{\beta}_1 x. \tag{10.1}$$

Step — (b) your friend's log odds. Dividing the softmax probability of orange by that of apple, the common denominator cancels:

$$\frac{p_o(x)}{p_a(x)} = \frac{\exp(\hat{\alpha}_{o0} + \hat{\alpha}_{o1} x)}{\exp(\hat{\alpha}_{a0} + \hat{\alpha}_{a1} x)}. \tag{10.2}$$

Taking logs of (10.2):

$$\log \frac{p_o(x)}{p_a(x)} = (\hat{\alpha}_{o0} - \hat{\alpha}_{a0}) + (\hat{\alpha}_{o1} - \hat{\alpha}_{a1}) x. \tag{10.3}$$

Step — (c) mapping your estimates to your friend's. Both models are fitted by maximum likelihood to the same data and (10.1) and (10.3) are the same family of functions, so the fitted log odds coincide for every x. Matching intercepts and slopes:

$$\hat{\alpha}_{o0} - \hat{\alpha}_{a0} = \hat{\beta}_0, \qquad \hat{\alpha}_{o1} - \hat{\alpha}_{a1} = \hat{\beta}_1. \tag{10.4}$$

Substituting $\hat{\beta}_0 = 2$, $\hat{\beta}_1 = -1$ into (10.4): $\hat{\alpha}_{o0} - \hat{\alpha}_{a0} = 2$ and $\hat{\alpha}_{o1} - \hat{\alpha}_{a1} = -1$. Only the differences are identified (adding the same constant to both classes' coefficients leaves every probability unchanged); with the common normalization $\hat{\alpha}_{a0} = \hat{\alpha}_{a1} = 0$, $\hat{\alpha}_{o0} = 2$ and $\hat{\alpha}_{o1} = -1$.

Step — (d) mapping your friend's estimates to yours. By (10.4) with $\hat{\alpha}_{o0} = 1.2$, $\hat{\alpha}_{o1} = -2$, $\hat{\alpha}_{a0} = 3$, $\hat{\alpha}_{a1} = 0.6$:

$$\hat{\beta}_0 = (1.2) - (3) = -1.8, \tag{10.5}$$

$$\hat{\beta}_1 = (-2) - (0.6) = -2.6. \tag{10.6}$$

Step — (e) agreement. The two fitted models give the same log odds (10.1)=(10.3) for every x, hence the same probabilities and the same labels under any common threshold. They agree on all 2,000 test observations: a fraction of 1 (100%). The parametrizations differ; the model does not.

Intuition. With two classes, softmax has one redundant set of coefficients: only the difference between the classes' linear predictors matters. Choosing a baseline class removes the redundancy without changing a single prediction.

Ref: ISLP §4.3.5, eqs. (4.10)–(4.13), pp. 144–146.

Grading rubric (10 points).

- (a) 1 pt; (b) with the cancellation shown: 2 pts.
- (c) the two difference equations and the identification remark: 3 pts.
- (d) $\hat{\beta}_0 = -1.8$, $\hat{\beta}_1 = -2.6$: 2 pts. (e) 100%, with the reason: 2 pts.
- Claiming in (c) that the friend's four coefficients are uniquely determined: max. 1.5 pts in (c).

Question 11. Classifying the direction of the market with `Weekly` (ISLP 4.13).

Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.13 (Section 4.8, Applied), pp. 196–197. Reproduced verbatim.

This question should be answered using the `Weekly` data set, which is part of the `ISLP` package. This data is similar in nature to the `Smarket` data from this chapter's lab, except that it contains 1,089 weekly returns for 21 years, from the beginning of 1990 to the end of 2010.

(a) Produce some numerical and graphical summaries of the `Weekly` data. Do there appear to be any patterns?

(b) Use the full data set to perform a logistic regression with `Direction` as the response and the five lag variables plus `Volume` as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?

(c) Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.

(d) Now fit the logistic regression model using a training data period from 1990 to 2008, with `Lag2` as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).

(e) Repeat (d) using LDA.

(f) Repeat (d) using QDA.

(g) Repeat (d) using KNN with $K = 1$.

(h) Repeat (d) using naive Bayes.

(i) Which of these methods appears to provide the best results on this data?

(j) Experiment with different combinations of predictors, including possible transformations and interactions, for each of the methods. Report the variables, method, and associated confusion matrix that appears to provide the best results on the held out data. Note that you should also experiment with values for K in the KNN classifier.

Notation.

- $Y_t \in \{0, 1\}$: `Direction` of week t (1 = Up); `Lag1`–`Lag5`: percentage returns of the previous five weeks; `Volume`: average daily shares traded (billions).

- Training set: 1990–2008 (985 weeks); test set: 2009–2010 (104 weeks).
- π_k: prior share of class k; μ_k, σ^2 (pooled) or σ_k^2: class mean and variance of `Lag2`; $\delta_k(x)$: discriminant score.
- Confusion matrices: rows = predicted (Down, Up), columns = true (Down, Up).

Step — (a) summaries. 1,089 weeks, of which a share 0.5556 went Up. The lag returns are almost uncorrelated with each other and with `Today` (largest absolute pairwise correlation 0.0757): past returns carry little linear information about future ones. The one strong pattern is `Volume` rising over time: its correlation with `Year` is 0.8419, and the mean volume goes from 0.1548 (1990) to 4.5578 (2010), so a plot of `Volume` against time rises steadily, while scatterplots of one lag against another are shapeless clouds, as the near-zero correlations imply.

Step — (b) full-sample logistic regression. Coefficients (p-values):

	const	`Lag1`	`Lag2`	`Lag3`	`Lag4`	`Lag5`	`Volume`
$\hat\beta$	0.2669	−0.0413	0.0584	−0.0161	−0.0278	−0.0145	−0.0227
p	0.0019	0.1181	0.0296	0.5469	0.2937	0.5833	0.5377

Only `Lag2` is significant at 5% (besides the intercept).

Step — (c) confusion matrix (training data). With the rule (4.2) of Question 4:

	true Down	true Up
predicted Down	54	48
predicted Up	430	557

The fraction correct is

$$\frac{54+557}{1089} = \frac{611}{1089} = 0.5611. \tag{11.1}$$

By true class: Up weeks are caught at a rate $557/(557+48) = 557/605 = 0.9207$, Down weeks at $54/(54+430) = 54/484 = 0.1116$. The model predicts Up almost always; its mistakes are mostly Down weeks called Up. Its training accuracy, 0.5611, barely beats always predicting Up (0.5556), and it is measured on the fitting data, so it is optimistic.

Step — (d) logistic regression, `Lag2`, held-out data. Training fit: $\hat\beta_0 = 0.2033$, $\hat\beta_1 = 0.0581$ ($p = 0.043$). The rule predicts Up when $\hat\beta_0 + \hat\beta_1 x > 0$, i.e. when

$$x > -\frac{\hat\beta_0}{\hat\beta_1} = -\frac{0.203257}{0.058095} = -3.4987. \tag{11.2}$$

On the 104 test weeks:

	true Down	true Up
predicted Down	9	5
predicted Up	34	56

Fraction correct: $(9+56)/104 = 65/104 = 0.625$.

Step — (e) LDA. With $p = 1$, the LDA discriminant (ISLP eq. 4.18) is

$$\delta_k(x) = x\,\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k. \tag{11.3}$$

LDA predicts Up iff $\delta_1(x) > \delta_0(x)$; subtracting the two versions of (11.3) and solving for x (with $\mu_1 > \mu_0$):

$$x > \frac{\mu_0+\mu_1}{2} - \frac{\sigma^2 \log(\pi_1/\pi_0)}{\mu_1-\mu_0}. \tag{11.4}$$

Training estimates: $\hat{\pi}_1 = 544/985 = 0.552284$, so $\log(\hat{\pi}_1/\hat{\pi}_0) = 0.209904$; $\hat{\mu}_0 = -0.035683$, $\hat{\mu}_1 = 0.260366$; pooled $\hat{\sigma}^2 = 5.132198$. Substituting into (11.4):

$$x > \frac{(-0.035683) + (0.260366)}{2} - \frac{(5.132198)(0.209904)}{(0.260366) - (-0.035683)} = 0.1123415 - \frac{1.077269}{0.296049} = -3.5265. \tag{11.5}$$

The threshold (11.5) is almost that of (11.2); no test week has `Lag2` between them, so the confusion matrix is identical to (d): accuracy 0.625.

Step — (f) QDA. QDA uses class variances, $\hat{\sigma}_0^2 = 4.8378$ and $\hat{\sigma}_1^2 = 5.3707$, in $\delta_k(x) = -\frac{(x-\mu_k)^2}{2\sigma_k^2} - \log \sigma_k + \log \pi_k$. On the test weeks it predicts Up every time:

	true Down	true Up
predicted Down	0	0
predicted Up	43	61

Fraction correct $61/104 = 0.5865$, the share of Up weeks in 2009–2010.

Step — (g) KNN, $K = 1$. Each test week takes the label of the training week with the closest `Lag2`:

	true Down	true Up
predicted Down	22	31
predicted Up	21	30

Fraction correct $(22 + 30)/104 = 52/104 = 0.500$: no better than a coin.

Step — (h) naive Bayes. With one predictor, Gaussian naive Bayes is QDA with variances computed with divisor n_k instead of $n_k - 1$; here the predictions are the same as in (f): all Up, accuracy 0.5865.

Step — (i) best method. Logistic regression and LDA (0.625), ahead of QDA and naive Bayes (0.5865, the no-skill rate) and KNN with $K = 1$ (0.500). The margin is small: with 104 test weeks the standard error of an accuracy near 0.625 is

$$\sqrt{\frac{0.625 \times 0.375}{104}} = \sqrt{0.0022536} = 0.0475, \tag{11.6}$$

so 0.625 is less than one standard error above 0.5865.

Step — (j) experiments. Held-out accuracy of other specifications:

predictors	logistic	LDA	QDA
`Lag2`	0.6250	0.6250	0.5865
`Lag1`+`Lag2`	0.5769	0.5769	0.5577
`Lag2`+`Lag2`^{2}	0.6250	0.6154	0.6250
`Lag1`–`Lag5`+`Volume`	0.4615	0.4615	0.4327

KNN on `Lag2`	accuracy
$K = 1$	0.5000
$K = 3$	0.5577
$K = 5$	0.5385
$K = 10$	0.5673
$K = 20$	0.5962
$K = 50$	0.5577
$K = 100$	0.5673
$K = 150$	0.6250

No specification beats 0.625. The best result is shared by logistic regression and LDA on `Lag2`, logistic regression and QDA on `Lag2` and its square, and KNN with $K = 150$; the simplest, logistic regression on `Lag2`, has the confusion matrix of (d). Adding predictors hurts (the full model falls below 0.5): extra noise variables add variance and no signal.

```
wk = pd.read_csv("data/Weekly.csv"); y = (wk.Direction == "Up").astype(int).to_numpy()
tr, te = (wk.Year <= 2008).to_numpy(), (wk.Year > 2008).to_numpy()
X = wk[["Lag2"]].to_numpy()
for fit in (logit_fit, lda_fit, qda_fit, lambda X, y: qda_fit(X, y, nb=True)):
    pred = (fit(X[tr], y[tr])(X[te]) > 0.5).astype(int); cm = confusion(pred, y[te])
pred_knn = knn_pred(X[tr], y[tr], X[te], K=1)   # nearest neighbour by |Lag2 difference|
```

Intuition. Weekly returns are close to unpredictable from their own past, so every classifier ends up near "always Up". The methods that win are the smooth, low-variance ones; flexible ones (KNN with small K, QDA, many predictors) fit noise in the training years.

Ref: ISLP §4.3 (logistic regression, pp. 138–146), §4.4.1–4.4.4 (LDA, QDA, naive Bayes, pp. 147–161), §4.5 (comparison, pp. 161–167), §2.2.3 (KNN, pp. 34–39).

Grading rubric (10 points).

- (a) Summaries with the volume pattern: 1 pt. (b) Fit and the significant predictor: 1 pt.
- (c) Matrix, accuracy and the type of mistakes: 1.5 pts.
- (d)–(h) Five held-out confusion matrices and accuracies: 1 pt each (5 pts).
- (i)–(j) Comparison and experiments, including K: 1.5 pts.
- Using the full sample for (d)–(h) instead of the 2009–2010 hold-out: max. 4 pts in the question.

Question 12. High versus low gas mileage with `Auto` (ISLP 4.14).
Source: James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*, Exercise 4.14 (Section 4.8, Applied), pp. 197–198. Reproduced verbatim.

In this problem, you will develop a model to predict whether a given car gets high or low gas mileage based on the `Auto` data set.

(a) Create a binary variable, `mpg01`, that contains a 1 if `mpg` contains a value above its median, and a 0 if `mpg` contains a value below its median. You can compute the median using the `median()` method of the data frame. Note you may find it helpful to add a column `mpg01` to the data frame by assignment. Assuming you have stored the data frame as `Auto`, this can be done as follows:

```
Auto['mpg01'] = mpg01
```

(b) Explore the data graphically in order to investigate the association between `mpg01` and the other features. Which of the other features seem most likely to be useful in predicting `mpg01`? Scatterplots and boxplots may be useful tools to answer this question. Describe your findings.

(c) Split the data into a training set and a test set.

(d) Perform LDA on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(e) Perform QDA on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(f) Perform logistic regression on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(g) Perform naive Bayes on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

(h) Perform KNN on the training data, with several values of K, in order to predict `mpg01`. Use only the variables that seemed most associated with `mpg01` in (b). What test errors do you obtain? Which value of K seems to perform the best on this data set?

Notation.

- $n = 392$ cars (the ISLP version of `Auto`, without the 5 cars missing `horsepower`); $Y_i = \texttt{mpg01} \in \{0, 1\}$.

- $\mathbf{x}_i \in \mathbb{R}^4$: (`cylinders`, `displacement`, `horsepower`, `weight`); $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}$ (pooled) or $\boldsymbol{\Sigma}_k$: class mean vector and covariance.
- Split: `default_rng(1).permutation(392)`; first 196 indices train, last 196 test.
- Err: test misclassification rate = (errors)/196.

Step — (a) the response. The median of `mpg` is 22.75 (the average of the two middle values; no car has exactly that value), so

$$Y_i = I(\texttt{mpg}_i > 22.75), \tag{12.1}$$

which puts 196 of the 392 cars in each class.

Step — (b) which features. Class means and the correlation with Y (a numerical version of the boxplots, which show the same separation):

feature	mean, $Y=0$	mean, $Y=1$	$\text{corr}(\cdot, Y)$
`cylinders`	6.77	4.18	−0.759
`displacement`	273.16	115.67	−0.753
`horsepower`	130.11	78.83	−0.667
`weight`	3620.40	2334.77	−0.758
`acceleration`	14.59	16.50	0.347
`year`	74.40	77.56	0.430
`origin`	1.16	1.99	0.514

The engine-size block (`cylinders`, `displacement`, `horsepower`, `weight`) separates the classes most ($|\text{corr}| \geq 0.667$; the boxplots barely overlap); these four are used below. They are strongly correlated with one another, so they carry overlapping information.

Step — (c) the split. 196 training cars (94 with $Y = 1$) and 196 test cars.

Step — (d)–(g) the four model-based classifiers. Each is trained on the training cars and classifies a test car to $Y = 1$ when its posterior probability exceeds 0.5; for LDA and QDA the posterior comes from Bayes' theorem with Gaussian class densities (ISLP eq. 4.15), for logistic regression from (4.1). Test errors:

method	errors	Err
(d) LDA	24	$24/196 = 0.1224$
(e) QDA	22	$22/196 = 0.1122$
(f) logistic regression	25	$25/196 = 0.1276$
(g) naive Bayes (Gaussian)	23	$23/196 = 0.1173$

Step — (h) KNN. The four predictors are on very different scales (`weight` in thousands, `cylinders` in units), so they are standardized with the training means and standard deviations before computing Euclidean distances. Test errors:

K	1	3	5	10	20	50	100
Err	0.1122	0.1020	0.1020	0.1173	0.1173	0.1224	0.1071

$K = 3$ and $K = 5$ do best ($20/196 = 0.1020$). All methods land between 0.10 and 0.13; with 196 test cars one error is $1/196 = 0.0051$, so the differences are a handful of cars and the ranking would change with another split.

```
auto = pd.read_csv("data/Auto.csv"); y = (auto.mpg > auto.mpg.median()).astype(int).to_numpy()
X = auto[["cylinders", "displacement", "horsepower", "weight"]].to_numpy(float)
idx = np.random.default_rng(1).permutation(len(y)); tr, te = idx[:196], idx[196:]
err = ((lda_fit(X[tr], y[tr])(X[te]) > 0.5) != y[te]).mean()
Z = (X - X[tr].mean(0)) / X[tr].std(0)              # KNN on standardized predictors
err_knn = (knn_pred(Z[tr], y[tr], Z[te], K=3) != y[te]).mean()
```

Intuition. When the classes are this well separated by a few strongly related variables, every reasonable classifier gets about 9 cars in 10 right; the remaining errors are cars near the median, where no method can be sure.

Ref: ISLP §4.4 (LDA, QDA, naive Bayes, pp. 146–161), §4.5 (comparison, pp. 161–167), §2.2.3 (KNN, pp. 34–39).

Grading rubric (10 points).

- (a) `mpg01` correctly built: 1 pt. (b) Features chosen with evidence: 2 pts. (c) Split: 1 pt.
- (d)–(g) Each test error: 1 pt (4 pts). (h) KNN for several K with the best value: 2 pts.
- Unstandardized KNN without comment on scale: -1 pt. Errors reported on the training set: max. 4 pts.

Block 2 — Wooldridge (2016), 6th ed. *(Ch. 17 • 3 problems)*

Question 13. Point spreads: linear probability model versus probit (W 17.C1).

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C1, p. 560. Reproduced verbatim.

Use the data in PNTSPRD for this exercise.

(i) The variable *favwin* is a binary variable if the team favored by the Las Vegas point spread wins. A linear probability model to estimate the probability that the favored team wins is

$$P(favwin = 1 \mid spread) = \beta_0 + \beta_1 spread.$$

Explain why, if the spread incorporates all relevant information, we expect $\beta_0 = .5$.

(ii) Estimate the model from part (i) by OLS. Test $H_0 \colon \beta_0 = .5$ against a two-sided alternative. Use both the usual and heteroskedasticity-robust standard errors.

(iii) Is *spread* statistically significant? What is the estimated probability that the favored team wins when $spread = 10$?

(iv) Now, estimate a probit model for $P(favwin = 1 \mid spread)$. Interpret and test the null hypothesis that the intercept is zero. [*Hint:* Remember that $\Phi(0) = .5$.]

(v) Use the probit model to estimate the probability that the favored team wins when $spread = 10$. Compare this with the LPM estimate from part (iii).

(vi) Add the variables *favhome*, *fav25*, and *und25* to the probit model and test joint significance of these variables using the likelihood ratio test. (How many df are in the chi-square distribution?) Interpret this result, focusing on the question of whether the spread incorporates all observable information prior to a game.

Notation.

- $favwin_i \in \{0, 1\}$: favored team wins game i; $spread_i \geq 0$: Las Vegas point spread (points by which the favorite is expected to win); $n = 553$ games.
- β_0, β_1: LPM (or probit) coefficients; Φ: standard normal cdf.
- *favhome*: favorite plays at home; *fav25*, *und25*: favorite / underdog in the top 25; ℓ_r, ℓ_{ur}: restricted and unrestricted log-likelihoods.

Step — (i) why $\beta_0 = .5$. If the spread summarizes all information, a game with $spread = 0$ has no favorite: each team wins with probability one half. In the LPM,

$$P(favwin = 1 \mid spread = 0) = \beta_0 + \beta_1(0) = \beta_0, \tag{13.1}$$

so (13.1) must equal .5.

Step — (ii) OLS and the test of $\beta_0 = .5$. OLS gives (usual standard errors in parentheses, heteroskedasticity-robust HC1 ones in brackets)

$$\widehat{favwin} = \underset{(0.0282)\ [0.0317]}{0.5769} + \underset{(0.0023)\ [0.0019]}{0.01937}\ spread, \qquad n = 553,\ R^2 = 0.1107. \tag{13.2}$$

The test statistic is

$$t = \frac{\hat{\beta}_0 - 0.5}{\mathrm{se}(\hat{\beta}_0)}. \tag{13.3}$$

With the usual standard error, substituting into (13.3):

$$t = \frac{0.5769493 - 0.5}{0.0282345} = \frac{0.0769493}{0.0282345} = 2.7254, \qquad p = 0.0066. \tag{13.4}$$

With the robust standard error:

$$t = \frac{0.0769493}{0.0316569} = 2.4307, \qquad p = 0.0151. \tag{13.5}$$

H_0 is rejected at 5% either way (at 1% only with the usual standard error). The LPM is heteroskedastic by construction ($\mathrm{Var}(y \mid x) = p(x)[1 - p(x)]$), so (13.5) is the appropriate test.

Step — (iii) significance of *spread* and the probability at 10. $t = 8.28$ (usual) and 10.08 (robust): highly significant. Substituting $spread = 10$ into (13.2):

$$\hat{P} = 0.576949 + 0.0193655\,(10) = 0.576949 + 0.193655 = 0.7706. \tag{13.6}$$

Step — (iv) probit. Maximum likelihood gives

$$\hat{P}(favwin = 1 \mid spread) = \Phi\big(\underset{(0.1037)}{-0.0106} + \underset{(0.0122)}{0.0925}\ spread\big). \tag{13.7}$$

At $spread = 0$ the probability is $\Phi(\beta_0)$, and $\Phi(\beta_0) = .5 \iff \beta_0 = 0$, so $H_0\colon \beta_0 = 0$ is the probit version of (i). The statistic is

$$z = \frac{-0.010593}{0.103747} = -0.1021, \qquad p = 0.919: \tag{13.8}$$

not rejected. Unlike the LPM, the probit does not contradict the efficient-spread hypothesis.

Step — (v) probit probability at 10. Substituting $spread = 10$ into (13.7):

$$\hat{P} = \Phi\big(-0.010593 + 0.092463\,(10)\big) = \Phi(-0.010593 + 0.924630) = \Phi(0.914037) = 0.8197. \tag{13.9}$$

It is about 5 percentage points above the LPM value (13.6), 0.7706. The LPM forces a straight line through 0.5769 at zero; the probit starts at $\Phi(-0.0106) \approx 0.5$ and curves up faster.

Step — (vi) likelihood ratio test. The LR statistic for the $q = 3$ exclusion restrictions (W eq. 17.12) is

$$LR = 2(\ell_{ur} - \ell_r) \overset{a}{\sim} \chi^2_3. \tag{13.10}$$

With $\ell_{ur} = -262.6418$ and $\ell_r = -263.5622$, substituting into (13.10):

$$LR = 2\big[(-262.6418) - (-263.5622)\big] = 2(0.9204) = 1.8408, \qquad p = 0.606. \tag{13.11}$$

The three variables are jointly (and individually: $p = 0.278,\ 0.985,\ 0.380$) insignificant. Home advantage and rankings are public before the game, and the spread already prices them in: evidence that the spread incorporates all observable information.

```
import wooldridge as wd
ps = wd.data("pntsprd"); X = sm.add_constant(ps[["spread"]])
ols = sm.OLS(ps.favwin, X).fit(); rob = sm.OLS(ps.favwin, X).fit(cov_type="HC1")
pr = sm.Probit(ps.favwin, X).fit(disp=0); p10 = stats.norm.cdf(pr.params @ [1, 10])
pr2 = sm.Probit(ps.favwin, sm.add_constant(ps[["spread", "favhome", "fav25", "und25"]])).fit(disp=0)

LR = 2 * (pr2.llf - pr.llf)
```

Intuition. Betting markets aggregate information; if they are efficient, the spread alone should predict outcomes and nothing public should add to it. The probit, which respects the $[0, 1]$ range, supports that; the LPM's rejection in (ii) comes partly from forcing a line.

Ref: W §17.1 (LPM, probit, LR test eq. 17.12, pp. 525–536); W §8.5 (LPM heteroskedasticity).

Grading rubric (10 points).

- (i) 1 pt. (ii) Estimates and both t tests: 2 pts. (iii) Significance and 0.7706: 1.5 pts.
- (iv) Probit, interpretation of $\beta_0 = 0$, test: 2 pts. (v) 0.8197 and comparison: 1.5 pts.
- (vi) LR, 3 df, conclusion about the spread: 2 pts.
- Testing $\beta_0 = 0.5$ instead of $\beta_0 = 0$ in the probit: max. 0.5 pt in (iv).

Question 14. Quadratics in the Poisson regression for arrests (W 17.C4).

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C4, p. 561. Reproduced verbatim.

In Example 9.1, we added the quadratic terms $pcnv^2$, $ptime86^2$, and $inc86^2$ to a linear model for *narr86*.

(i) Use the data in CRIME1 to add these same terms to the Poisson regression in Example 17.3.

(ii) Compute the estimate of σ^2 given by $\hat{\sigma}^2 = (n - k - 1)^{-1} \sum_{i=1}^{n} \hat{u}_i^2 / \hat{y}_i$. Is there evidence of overdispersion? How should the Poisson MLE standard errors be adjusted?

(iii) Use the results from parts (i) and (ii) and Table 17.5 to compute the quasi-likelihood ratio statistic for joint significance of the three quadratic terms. What do you conclude?

Notation.

- $y_i = narr86_i \in \{0, 1, 2, \dots\}$: arrests in 1986; $n = 2725$ men.
- $\mathbf{x}_i$: *pcnv* (share of prior arrests leading to conviction), *avgsen*, *tottime* (months), *ptime86* (months in prison in 1986), *qemp86* (quarters employed), *inc86* (hundreds of USD), *black*, *hispan*, *born60*, plus the three squares; $k = 12$ slopes.
- $\hat{y}_i = \exp(\mathbf{x}_i \hat{\boldsymbol{\beta}})$; $\hat{u}_i = y_i - \hat{y}_i$; $\sigma^2 > 0$: dispersion factor in $\mathrm{Var}(y \mid \mathbf{x}) = \sigma^2 E(y \mid \mathbf{x})$ (W eq. 17.35).

Step — (i) the Poisson regression with quadratics. The model is $E(y \mid \mathbf{x}) = \exp(\mathbf{x}\boldsymbol{\beta})$ (W eq. 17.31), estimated by Poisson (Q)MLE:

variable	$\hat{\beta}$	Poisson s.e.	adjusted s.e. ($\hat{\sigma}\times$ s.e.)
constant	−0.7099	0.0702	0.0828
pcnv	1.1531	0.2823	0.3329
avgsen	−0.0256	0.0215	0.0253
tottime	0.0122	0.0158	0.0186
ptime86	0.6837	0.0912	0.1076
qemp86	0.0230	0.0327	0.0386
inc86	−0.01207	0.00183	0.00215
black	0.5914	0.0744	0.0877
hispan	0.4220	0.0748	0.0883
born60	−0.0929	0.0642	0.0758
$pcnv^2$	−1.7951	0.3072	0.3623
$ptime86^2$	−0.1034	0.0162	0.0191
$inc86^2$ ($\times 10^5$)	2.068	0.614	0.724
log-likelihood	$\ell_{ur} = -2168.866$		

All three squares are individually significant. Each linear-plus-square pair has a turning point at $-\hat{\beta}_x / (2\hat{\beta}_{x^2})$; e.g. for *pcnv*:

$$pcnv^* = \frac{1.153133}{2(1.795063)} = \frac{1.153133}{3.590126} = 0.3212, \tag{14.1}$$

so expected arrests rise with the conviction rate up to 0.32 and fall after it. Likewise *ptime86* peaks at $0.683681/(2 \times 0.103440) = 0.683681/0.206880 = 3.3047$ months, while *inc86* has a minimum at about 292 (i.e. 29,200 dollars), beyond which the data are thin.

Step — (ii) dispersion. The dispersion estimator (W, p. 546) uses the $n-k-1$ degrees of freedom:

$$\hat{\sigma}^2 = \frac{1}{n-k-1}\sum_{i=1}^{n}\frac{\hat{u}_i^2}{\hat{y}_i} = \frac{3772.81}{2725-12-1} = \frac{3772.81}{2712} = 1.3912, \tag{14.2}$$

and $\hat{\sigma} = \sqrt{1.3912} = 1.1795$. Since $\hat{\sigma}^2 > 1$ the variance exceeds the mean: overdispersion (milder than without the squares, where the same formula gives $\hat{\sigma} = 1.2316$, the 1.232 of Table 17.5). Under (14.2)'s variance assumption every Poisson MLE standard error is multiplied by $\hat{\sigma} = 1.1795$ (last column of the table), about 18% larger.

Step — (iii) quasi-likelihood ratio. The quasi-LR statistic divides the usual LR statistic by $\hat{\sigma}^2$ from the unrestricted model (W, p. 546):

$$QLR = \frac{2(\ell_{ur} - \ell_r)}{\hat{\sigma}^2} \overset{a}{\sim} \chi^2_3. \tag{14.3}$$

The restricted model is that of Table 17.5, $\ell_r = -2248.761$ (our re-estimate reproduces the table's $-2{,}248.76$). Substituting it, ℓ_{ur} from (i) and (14.2) into (14.3):

$$QLR = \frac{2\big[(-2168.866) - (-2248.761)\big]}{1.3912} = \frac{2(79.895)}{1.3912} = \frac{159.789}{1.3912} = 114.86. \tag{14.4}$$

The 5% critical value of χ^2_3 is 7.81; $p < 0.001$. The quadratic terms are jointly very significant even after the overdispersion correction (the uncorrected LR would be 159.79).

```
cr = wd.data("crime1"); pois = sm.families.Poisson()
base = ["pcnv","avgsen","tottime","ptime86","qemp86","inc86","black","hispan","born60"]
ur = sm.GLM(cr.narr86, sm.add_constant(cr[base + ["pcnvsq","pt86sq","inc86sq"]]), family=pois).fit()

r = sm.GLM(cr.narr86, sm.add_constant(cr[base]), family=pois).fit()      # Table 17.5
s2 = np.sum((cr.narr86 - ur.predict())**2 / ur.predict()) / (len(cr) - 12 - 1)
QLR = 2 * (ur.llf - r.llf) / s2
```

Intuition. The Poisson mean is robust to a wrong variance, but its standard errors and LR tests are not; overdispersion means the data are noisier than Poisson, so each test statistic must be deflated by $\hat{\sigma}^2$. Here the signal in the quadratics is strong enough to survive.

Ref: W §17.3 (Poisson regression, QMLE, eqs. 17.31–17.35, quasi-LR, Example 17.3 and Table 17.5, pp. 543–547); W Example 9.1 (p. 275).

Grading rubric (10 points).

- (i) Poisson fit with the three squares: 3 pts.
- (ii) $\hat{\sigma}^2$ with the right df, overdispersion verdict, the $\hat{\sigma}$ adjustment: 3 pts.
- (iii) QLR with $\hat{\sigma}^2$ in the denominator, χ^2_3, conclusion: 4 pts.
- Using the unadjusted LR as the answer to (iii): max. 2 pts in (iii).

Question 15. Poisson regression for fertility (W 17.C5).

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 17, Computer Exercise C5, p. 561. Reproduced verbatim.

Refer to Table 13.1 in Chapter 13. There, we used the data in FERTIL1 to estimate a linear model for *kids*, the number of children ever born to a woman.

(i) Estimate a Poisson regression model for *kids*, using the same variables in Table 13.1. Interpret the coefficient on *y82*.

(ii) What is the estimated percentage difference in fertility between a black woman and a nonblack woman, holding other factors fixed?

(iii) Obtain $\hat{\sigma}$. Is there evidence of over- or underdispersion?

(iv) Compute the fitted values from the Poisson regression and obtain the R-squared as the squared correlation between $kids_i$ and $\widehat{kids}_i$. Compare this with the R-squared for the linear regression model.

Notation.

- $y_i = kids_i$: children ever born; $n = 1129$ women (GSS, even years 1972–1984).
- Regressors of Table 13.1 ($k = 17$): *educ*, *age*, age^2, *black*, region at 16 (*east*, *northcen*, *west*; base South), living environment at 16 (*farm*, *othrural*, *town*, *smcity*; base large city), year dummies *y74*–*y84* (base 1972).
- $\hat{y}_i = \exp(\mathbf{x}_i\hat{\boldsymbol{\beta}})$; $\hat{\sigma}^2$ as in (14.2).

Step — (i) estimates and *y82*. Poisson QMLE (standard errors):

variable	$\hat{\beta}$	s.e.	variable	$\hat{\beta}$	s.e.
constant	−3.0605	1.2107	*othrural*	−0.0573	0.0692
educ	−0.0482	0.0072	*town*	0.0307	0.0486
age	0.2045	0.0548	*smcity*	0.0741	0.0615
age^2	−0.00223	0.00062	*y74*	0.0933	0.0631
black	0.3603	0.0611	*y76*	−0.0288	0.0676
east	0.0878	0.0527	*y78*	−0.0157	0.0687
northcen	0.1417	0.0475	*y80*	−0.0197	0.0690
west	0.0795	0.0657	*y82*	−0.1926	0.0675
farm	−0.0148	0.0576	*y84*	−0.2144	0.0695

With an exponential mean, a dummy's exact proportional effect is

$$\%\Delta E(y \mid \mathbf{x}) = 100\big[\exp(\hat{\beta}_j) - 1\big]. \tag{15.1}$$

For *y82*, substituting into (15.1):

$$100\big[\exp(-0.192608) - 1\big] = 100\,(0.824806 - 1) = -17.52. \tag{15.2}$$

Holding education, age, race, region and environment fixed, women in 1982 had about 17.5% fewer children than in 1972 ($100\hat{\beta} = -19.3\%$ is the log approximation). The linear model's −0.522 children is, relative to the mean of 2.743, a similar −19%.

Step — (ii) black versus nonblack. Substituting $\hat{\beta}_{black} = 0.360348$ into (15.1):

$$100\big[\exp(0.360348) - 1\big] = 100\,(1.433828 - 1) = 43.38: \tag{15.3}$$

a black woman is estimated to have about 43% more children, other factors fixed.

Step — (iii) dispersion. With $k = 17$ slopes, the estimator (14.2) gives

$$\hat{\sigma}^2 = \frac{990.43}{1129 - 17 - 1} = \frac{990.43}{1111} = 0.8915, \qquad \hat{\sigma} = \sqrt{0.8915} = 0.9442. \tag{15.4}$$

$\hat{\sigma} < 1$: mild underdispersion. The usual Poisson standard errors are slightly too large (conservative); multiplying them by 0.9442 shrinks them by about 6%.

Step — (iv) goodness of fit. The squared correlation between $kids_i$ and $\hat{y}_i$ is 0.1209, against $R^2 = 0.1295$ for OLS (Table 13.1). OLS maximizes R^2 by construction, so it must be at least as high; the Poisson fit is close, while also respecting $\hat{y}_i > 0$ (and its fitted values average exactly $\bar{y} = 2.743$).

```
fe = wd.data("fertil1")
v = ["educ","age","agesq","black","east","northcen","west","farm","othrural","town",
     "smcity","y74","y76","y78","y80","y82","y84"]
po = sm.GLM(fe.kids, sm.add_constant(fe[v]), family=sm.families.Poisson()).fit()
pct_y82 = 100 * (np.exp(po.params.y82) - 1); yh = po.predict()
s2 = np.sum((fe.kids - yh)**2 / yh) / (len(fe) - len(v) - 1); r2 = np.corrcoef(fe.kids, yh)[0,
    1]**2
```

Intuition. Counts like the number of children are small non-negative integers; the exponential mean turns each coefficient into a proportional effect, which is the natural way to compare fertility across groups and years.

Ref: W §17.3 (Poisson regression, pp. 543–547); W Example 13.1 and Table 13.1 (p. 404).

Grading rubric (10 points).

- (i) Poisson fit and the *y82* effect as a percentage: 3 pts.
- (ii) 43.4% (or 36.0% with the log approximation, stated as such): 2 pts.
- (iii) $\hat{\sigma}$ and the underdispersion verdict: 2.5 pts. (iv) Both R^2 and the comparison: 2.5 pts.
- Reading the Poisson coefficient on *y82* as -0.19 children: max. 1 pt in (i).

Block 3 — Morettin & Singer (2021) *(Ch. 8 • 4 problems (in Portuguese))*

Question 16. KNN with LOOCV for the disc data (MS 8.9).

Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 9 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original).

Use o conjunto de dados do Exemplo 8.1 e distância aberta como variável preditora. Use LOOCV e o classificador KNN, com vizinhos mais próximos de 1 a 5.

a) Qual o melhor classificador baseado na acurácia?

b) Obtenha a matriz de confusão e realize o teste de McNemar.

c) Obtenha a sensibilidade e a especificidade e explique seus significados nesse caso.

Notation.

- Data of MS Example 8.1 (file `disco`): $n = 104$ temporomandibular joints, $y_i = 1$ if the disc is displaced (29 joints), x_i: open-mouth capsule–condyle distance (mm), recorded to 0.1 mm.
- $K \in \{1, \ldots, 5\}$: number of neighbours; V_0: the neighbour set of a point.
- $n_{11}, n_{12}, n_{21}, n_{22}$: cells of MS Tabela 8.2 (rows: decision sim/não; columns: true sim/não).

Step — the classifier. For a joint with distance x_0, KNN estimates (MS eq. 8.27)

$$\hat{P}(Y = 1 \mid X = x_0) = \frac{1}{K} \sum_{i \in V_0} I(y_i = 1), \tag{16.1}$$

and classifies to "displaced" when (16.1) exceeds 0.5. Under LOOCV the joint itself is left out of the candidate neighbours. Because distances are recorded to 0.1 mm, many neighbours are tied; as in R's `class::knn`, every point tied with the K-th distance votes (so V_0 can have more than K points), and an exactly split vote goes to the majority class "not displaced".

Step — a) best K. LOOCV accuracies:

K	1	2	3	4	5
accuracy	0.8173	0.8173	0.7885	0.8462	0.8077

$K = 4$ is best, with $88/104 = 0.8462$ correct, about the 84–85% of the logistic classifiers of MS Tabela 8.6.

Step — b) confusion matrix and McNemar. LOOCV predictions with $K = 4$:

	real: sim	real: não
decisão: sim	$n_{11} = 15$	$n_{12} = 2$
decisão: não	$n_{21} = 14$	$n_{22} = 73$

McNemar's statistic (MS eq. 8.35) tests H_0: $p_{12} = p_{21}$, i.e. that the classifier predicts displacement as often as it occurs:

$$M = \frac{(n_{12} - n_{21})^2}{n_{12} + n_{21}}. \tag{16.2}$$

Substituting the off-diagonal cells into (16.2):

$$M = \frac{(2 - 14)^2}{2 + 14} = \frac{144}{16} = 9.00, \qquad p = 0.0027 \ (\chi^2_1). \tag{16.3}$$

Since $n_{12} + n_{21} = 16 < 25$, MS recommend the exact binomial test: $p = 0.0042$. Both reject H_0: the classifier calls "displaced" $15 + 2 = 17$ times against 29 real cases, i.e. it under-predicts displacement.

Step — c) sensitivity and specificity. From MS Tabela 8.2,

$$\text{sensitivity} = \frac{n_{11}}{n_{11} + n_{21}} = \frac{15}{15 + 14} = \frac{15}{29} = 0.5172, \tag{16.4}$$

$$\text{specificity} = \frac{n_{22}}{n_{12} + n_{22}} = \frac{73}{2 + 73} = \frac{73}{75} = 0.9733. \tag{16.5}$$

By (16.4), the ultrasound rule detects only 52% of the discs that magnetic resonance shows to be displaced; by (16.5), it clears 97% of the healthy joints. For a screening test that replaces MRI, the missed displacements are the costly error: the majority vote (threshold 0.5) favours the larger class. A lower vote threshold, or the logistic cut-off of MS Example 8.1 (sensitivity 83%, specificity 84%), balances the two.

```
disco = pd.read_excel("data/disco.xls", sheet_name="dados")
x, y = disco.distanciaA.to_numpy(), disco.deslocamento.to_numpy()
def knn_loocv_ties(x, y, K):
    pred = np.empty(len(y), dtype=int)
    for i in range(len(y)):
        d = np.abs(np.delete(x, i) - x[i]); yy = np.delete(y, i)
        share = yy[d <= np.sort(d)[K - 1] + 1e-12].mean()   # all ties at the K-th distance vote
        pred[i] = int(share > 0.5)                           # a split vote goes to class 0
    return pred
```

Intuition. Accuracy rewards the majority class: with 72% of joints healthy, a classifier can be 84% accurate while missing half of the displaced discs. McNemar's test and the sensitivity/specificity pair expose that imbalance.

Ref: MS §8.4 (KNN, pp. 282–284), §8.6 Notas 1–2 (LOOCV and McNemar's test, pp. 285–288), Example 8.1 and Tabela 8.2 (pp. 264–268); AME Ch. 7 (confusion matrix, pp. 133–139).

Grading rubric (10 points).

- a) LOOCV accuracies for $K = 1, \ldots, 5$ and the best K: 4 pts.
- b) Confusion matrix and McNemar's statistic with its conclusion: 3 pts.
- c) Sensitivity, specificity and their clinical meaning: 3 pts.
- Using resubstitution (the joint among its own neighbours) instead of LOOCV: max. 4 pts.

Question 17. LDA with two inhibin predictors and k-fold CV (MS 8.4).

Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 4 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original; "varáveis" is the book's spelling).

Reanalise os dados do Exemplo 8.3 considerando duas varáveis preditoras `inibpre` e `inibpos` e validação cruzada de ordem k com diferentes valores de k. Avalie o efeito da semente nos resultados.

Notation.

- Data of MS Example 8.3 (file `inibina`): $n = 32$ patients, $y_i = 1$ for a positive response (19 patients), $\mathbf{x}_i = (\texttt{inibpre}_i, \texttt{inibpos}_i)'$, inhibin before and after stimulation.
- $\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_0$: class mean vectors; $\mathbf{S}_1, \mathbf{S}_0$: class covariance matrices; $\mathbf{S}$: pooled covariance; $\hat{\pi}_1 = n_1/n$.
- $k \in \{2, 4, 8, 16, 32\}$: number of CV folds ($k = 32$ is LOOCV); seeds $1, \ldots, 5$ of `default_rng`, which assign the folds.

Step — the classifier. Example 8.3 uses the Bayes classifier with Gaussian classes and a common covariance (`lda()`, priors equal to the sample shares). With two predictors the pooled covariance is (MS eq. 8.17)

$$\mathbf{S} = \frac{(n_1 - 1)\mathbf{S}_1 + (n_0 - 1)\mathbf{S}_0}{n_1 + n_0 - 2}, \tag{17.1}$$

and a patient is classified as positive when (MS eqs. 8.11 and 8.18 with priors)

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_0)'\mathbf{S}^{-1}\mathbf{x} \;\geq\; \tfrac{1}{2}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_0)'\mathbf{S}^{-1}(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_0) + \log(\hat{\pi}_0/\hat{\pi}_1). \tag{17.2}$$

Fitted to all 32 patients and evaluated on them (resubstitution), (17.2) is correct for $25/32 = 0.78125$ of them; with the single predictor $\texttt{difinib} = \texttt{inibpos} - \texttt{inibpre}$ of Example 8.3 the same computation gives $26/32 = 0.8125$, the book's 81%.

Step — k-fold CV. For each k and seed, the patients are split at random into k folds; (17.1)–(17.2) are re-estimated without each fold and applied to it; the accuracy is the share of the 32 held-out predictions that are correct (MS eq. 8.34 with accuracy in place of error). Number of correct predictions out of 32:

k	seed 1	seed 2	seed 3	seed 4	seed 5	mean accuracy	range (correct)
2	25	21	24	19	22	0.69375	6
4	25	24	21	22	22	0.7125	4
8	24	23	23	24	22	0.725	2
16	24	21	23	22	22	0.700	3
32 (LOOCV)	22	22	22	22	22	0.6875	0

For example, the mean for $k = 2$ is $(25 + 21 + 24 + 19 + 22)/(5 \times 32) = 111/160 = 0.69375$.

Step — effect of the seed. The seed matters a lot when k is small: with $k = 2$ the accuracy goes from $19/32 = 0.59375$ to $25/32 = 0.78125$, a range of $6/32 = 0.1875$, because each model is trained on only 16 patients and which 16 they are changes the rule. The range shrinks as k grows and is zero for LOOCV, which involves no randomness. All CV accuracies (about 0.69–0.73) are below the resubstitution 0.78: evaluating a rule on the data that built it is optimistic. With $n = 32$ one patient is worth $1/32 = 0.03125$ of accuracy, so no k is clearly better than another; the two predictors do not beat `difinib` alone.

```
ini = pd.read_excel("data/inibina.xls", sheet_name="dados")
y = (ini.resposta == "positiva").astype(int).to_numpy(); X = ini[["inibpre", "inibpos"]].to_numpy(
    float)
for k in (2, 4, 8, 16, 32):
    for s in (1, 2, 3, 4, 5):
        pred, acc = cv_accuracy(lda_fit, X, y, k, s)   # folds: default_rng(s).permutation(n) % k
```

Intuition. Cross-validation is itself an estimate with sampling noise. With tiny samples the random fold assignment is a large part of that noise; repeating CV over seeds (MS's "VC5/5") or using LOOCV removes the dependence on one split, though not the small-sample uncertainty.

Ref: MS §8.3.1 (Bayes classifier and LDA, eqs. 8.11, 8.17, 8.18, pp. 272–277), Example 8.3 (pp. 274–276), §8.6 Nota 1 (cross-validation, pp. 285–287); AME §1.5.1 (pp. 14–21).

Grading rubric (10 points).

- LDA with both predictors correctly specified: 3 pts.
- k-fold CV for at least three values of k: 3 pts.
- Seed effect measured over several seeds and explained (training size, LOOCV deterministic): 4 pts.
- Reporting only resubstitution accuracy: max. 3 pts.

Question 18. Choosing a logistic model for endometriosis by 5-fold CV (MS 8.6).
Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 6 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original).

Considere os dados do arquivo `endometriose2`. Com propósitos de previsão, ajuste modelos de regressão logística com validação cruzada de ordem 5 tendo `endometriose` como variável resposta e

i) `idade`, `dormenstrual`, `dismenorreia` ou `tipoesteril` separadamente como variáveis explicativas (4 modelos);

ii) cada par das variáveis do item i) como variáveis explicativas (6 modelos);

iii) cada trio das variáveis do item i) como variáveis explicativas (4 modelos);

iv) as quatro variáveis do item i) como variáveis explicativas (1 modelo).

Escolha o melhor modelo com base na acurácia.

Notation.

- $y_i = 1$ if patient i has endometriosis. The file has 1872 patients; dropping one with missing `idade`, one with missing `dismenorreia` and the five whose `tipoesteril` is "sim" without a type (one patient is in two of these groups) leaves $n = 1866$, of whom 492 have endometriosis.
- Coding: `idade` in years; `dormenstrual` a dummy (sim = 1); `dismenorreia` four dummies (leve, moderada, intensa, incapacitante; base "não"); `tipoesteril` two dummies (primária, secundária; base "não").
- Accuracy: share of correct held-out predictions with threshold 0.5; folds from `default_rng(1)`, the same for all 15 models; "VC5/5": mean over seeds $1, \ldots, 5$.

Step — models and criterion. Each model is the logistic regression (4.1) (MS §8.2) with the chosen predictors; its 5-fold CV accuracy is

$$\text{Acc} = 1 - \frac{1}{n}\sum_{i=1}^{n} I(y_i \neq \hat{y}_i^{(-j(i))}), \tag{18.1}$$

where $\hat{y}_i^{(-j(i))}$ is the prediction from the model fitted without the fold $j(i)$ that contains i (MS eq. 8.34). The benchmark is the rule "never endometriosis", with accuracy $1 - 492/1866 = 0.7363$.

Step — results. 5-fold CV accuracy (seed 1) and VC5/5:

item	predictors	5-fold	VC5/5
i)	`idade`	0.7363	0.7363
	`dormenstrual`	0.7363	0.7363
	`dismenorreia`	0.7358	0.7277
	`tipoesteril`	0.7497	0.7497
ii)	`idade+dormenstrual`	0.7363	0.7363
	`idade+dismenorreia`	0.7476	0.7447
	`idade+tipoesteril`	0.7460	0.7468
	`dormenstrual+dismenorreia`	0.7353	0.7273
	`dormenstrual+tipoesteril`	0.7588	0.7588
	`dismenorreia+tipoesteril`	**0.7685**	**0.7685**
iii)	`idade+dormenstrual+dismenorreia`	0.7481	0.7448
	`idade+dormenstrual+tipoesteril`	0.7588	0.7588
	`idade+dismenorreia+tipoesteril`	0.7503	0.7559
	`dormenstrual+dismenorreia+tipoesteril`	0.7680	0.7681
iv)	all four	0.7503	0.7550

The best model is `dismenorreia` + `tipoesteril`, with accuracy 0.7685 under both criteria, 3.2 points above the benchmark ($0.7685 - 0.7363 = 0.0322$). Models with categorical predictors only give the same predictions in every fold (they classify whole cells), which is why several entries coincide.

Step — reading the best model. Fitted to all 1866 patients, the log odds are -2.4112 (base: no dysmenorrhoea, not sterile) plus 0.2264 (leve), 2.1984 (moderada), 2.1561 (intensa), 1.5647 (incapacitante), and 1.3996 (primary sterility) or -0.0655 (secondary). A patient is classified as positive when the log odds are positive. For moderate dysmenorrhoea without sterility, $-2.4112 + 2.1984 = -0.2128 < 0$ (negative); with primary sterility,

$$-2.4112 + 2.1984 + 1.3996 = 1.1868 > 0 \tag{18.2}$$

(positive, probability 0.766). Only the cells "moderate, intense or incapacitating dysmenorrhoea and primary sterility" are classified as endometriosis (132 patients).

Step — `dormenstrual` adds nothing. Menstrual pain (`dormenstrual`) and dysmenorrhoea are almost the same variable: in only 3 of the 1872 patients does "no menstrual pain" disagree with "no dysmenorrhoea". Adding one to the other cannot improve the classifier; the near-collinearity also makes the logistic fit unstable in some folds (handled by IRLS with a pseudo-inverse).

```
end = pd.read_excel("data/endometriose2.xls", sheet_name="dados")
end = end.dropna(subset=["idade", "dismenorreia"]); end = end[end.tipoesteril != "sim"]
blocks = {"idade": end[["idade"]], "dormenstrual": end[["dormenstrual"]] == "sim",
          "dismenorreia": pd.get_dummies(end.dismenorreia)[["leve","moderada","intensa","
    incapacitante"]],
          "tipoesteril": pd.get_dummies(end.tipoesteril)[["primaria","secundaria"]]}
for m in [c for r in range(1, 5) for c in combinations(blocks, r)]:          # the 15 models
    X = np.column_stack([np.asarray(blocks[b], float) for b in m])
    acc = cv_accuracy(logit_fit, X, y, 5, seed=1)[1]
```

Intuition. Accuracy is a blunt criterion when one class is 74% of the data: most models barely beat "always no". What distinguishes the winner is that the combination of painful periods and primary sterility singles out a group in which endometriosis is the majority.

Ref: MS §8.2 (logistic classification and CV, pp. 264–271), §8.6 Nota 1 (pp. 285–287); MS §6.5 (logistic regression).

Grading rubric (10 points).

- Correct coding of the categorical predictors (dummies): 2 pts.
- The 15 models with 5-fold CV accuracy: 5 pts.
- Choice of the best model, compared with the no-information rate: 3 pts.
- Treating `dismenorreia` as a numeric score without saying so: -1 pt.

Question 19. Logistic regression, Bayes classifier and Fisher's discriminant (MS 8.11).

Source: Morettin & Singer (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), Capítulo 8, Exercício 11 (Seção 8.7, Exercícios), p. 289. Reproduced verbatim (Portuguese original). Exercise 5, to which it refers, is quoted for context.

Para os dados do Exercício 5, compare os classificadores usando regressão logística, classificador bayesiano e função discriminante linear de Fisher.

Exercise 5 (p. 289): Analise os dados do arquivo `endometriose2` por meio das técnicas de classificação descritas neste capítulo, utilizando validação cruzada com diferentes parâmetros. Compare-as e discuta os resultados.

Notation.

- Data and coding as in Question 18 ($n = 1866$, 492 positives); $\mathbf{x}_i \in \mathbb{R}^8$: the four predictors of model iv (`idade`, `dormenstrual`, 4 `dismenorreia` dummies, 2 `tipoesteril` dummies).
- $\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_0$, $\mathbf{S}$: class means and pooled covariance (17.1); $\hat{\pi}_1 = 492/1866$.
- Evaluation: 5-fold CV repeated with seeds $1, \ldots, 5$ (VC5/5); sensitivity = share of true positives classified positive; specificity = share of true negatives classified negative.

Step — the three classifiers. (1) Logistic regression (4.1), positive if $\hat{p}(\mathbf{x}) > 0.5$. (2) The Bayes classifier of MS §8.3.1 with Gaussian classes and common covariance (LDA), positive if rule (17.2) holds, which includes the prior term $\log(\hat{\pi}_0/\hat{\pi}_1)$. (3) Fisher's linear discriminant (MS eq. 8.18 and Example 8.6), positive if

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_0)'\mathbf{S}^{-1}\mathbf{x} \;\geq\; \tfrac{1}{2}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_0)'\mathbf{S}^{-1}(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_0), \tag{19.1}$$

i.e. the same direction as (2) with equal priors. Since $\hat{\pi}_1 = 0.264 < 0.5$, the prior term $\log(\hat{\pi}_0/\hat{\pi}_1) > 0$ raises the threshold in (2); (19.1) drops it, so Fisher's rule calls more patients positive.

Step — VC5/5 results.

classifier	accuracy	sensitivity	specificity
logistic regression	0.7550	0.2337	0.9416
Bayes classifier (LDA, sample priors)	0.7534	0.2854	0.9210
Fisher's discriminant (equal priors)	0.6980	0.7646	0.6741

Logistic regression and the Bayes classifier are nearly the same rule (their first-repetition CV predictions agree for 97.5% of the patients). Fisher's rule loses about 5.7 points of accuracy but detects more than three times as many endometriosis cases (0.7646 against 0.2337).

Step — McNemar comparisons. On the first repetition, for each pair, n_{12} counts patients classified correctly by the first rule only and n_{21} by the second only (MS Nota 3, Tabela 8.8); the statistic is (16.2) of Question 16:

$$\text{logistic vs. Bayes:}\quad M = \frac{(19-28)^2}{19+28} = \frac{81}{47} = 1.7234, \quad p = 0.189, \tag{19.2}$$

$$\text{logistic vs. Fisher:}\quad M = \frac{(365-270)^2}{365+270} = \frac{9025}{635} = 14.2126, \quad p = 0.00016, \tag{19.3}$$

$$\text{Bayes vs. Fisher:}\quad M = \frac{(346-242)^2}{346+242} = \frac{10816}{588} = 18.3946, \quad p = 0.000018. \tag{19.4}$$

By (19.2) logistic regression and the Bayes classifier are not distinguishable; by (19.3)–(19.4) both are more accurate than Fisher's rule.

Step — which to use. If all errors cost the same, logistic regression or the Bayes classifier (accuracy about 0.755, barely above the 0.736 of "always no"). In a clinical screening, where missing an endometriosis case is worse than a false alarm, Fisher's rule, or either of the others with a lower probability threshold, is preferable: accuracy is the wrong yardstick there. The Gaussian assumption behind (2) and (3) is false for dummy predictors, yet the rules still work because they only need a good linear direction.

```
X = np.column_stack([np.asarray(blocks[b], float) for b in blocks])     # model iv of Question 18
clfs = {"logit": logit_fit, "bayes": lda_fit,
        "fisher": lambda X, y: lda_fit(X, y, prior=(0.5, 0.5))}          # (8.18): equal priors
for name, f in clfs.items():
    for s in (1, 2, 3, 4, 5):
        pred, acc = cv_accuracy(f, X, y, 5, s)
```

Intuition. The three rules share the same linear score up to the threshold; what really separates them is where the threshold sits, i.e. how much weight the minority class gets. Changing priors moves the classifier along the sensitivity–specificity trade-off, not to a better or worse score.

Ref: MS §8.2–8.3 (logistic regression, Bayes classifier, Fisher's discriminant, Example 8.6, pp. 264–282), §8.6 Notas 2–3 (McNemar, comparison of classifiers, pp. 287–288); ISLP §4.4.2 (LDA and the threshold, pp. 150–156).

Grading rubric (10 points).

- The three classifiers correctly specified (priors in Bayes, equal weights in Fisher): 3 pts.
- Cross-validated accuracy, sensitivity and specificity: 4 pts.
- A formal comparison (McNemar) and a recommendation tied to the costs of errors: 3 pts.
- Comparing resubstitution accuracies only: max. 4 pts.